

ASTE-331a: Spacecraft Systems Engineering

Propulsion System

(aka Propulsion or simply Prop)

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September 5th, 2025

Overview

Propulsion

Introduction

• Description

• Examples

Background

Application

Hardware

Integrated System

Additional Info

• Function

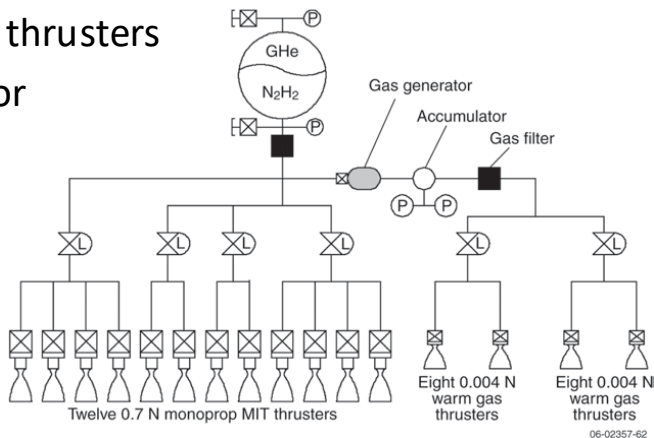
- Provides thrust for both trajectory design maneuvers and (often) attitude control
- The primary driver is typically the trajectory and corresponding ΔV budget.
- Across all subsystems, components are typically selected as those that meet the minimum requirements with the lowest mass, power, & cost.

• Common Components

- Chemical Propulsion (Mono/Biprop)
 - Fuel tanks
 - Thrusters (often different types tailored for small/large maneuvers)
 - Integration/etc. hardware
- Solid rocket motors (SRMs)
- Electric Propulsion Systems

• Example: Small Solar Orbiter

- 12 0.7 N orbit & attitude control thrusters
- 8 0.04 N warm gas thrusters
- Warm gas generator
- Propellant tank
- Misc. hardware
- Total = ~13 kg



• Key Trades & Analyses

- Propulsion type (solid, mono/bi, electric)
- Thruster & fuel tank sizing
- Heritage from prior systems

• Key Parameters

- Subsystem mass, power, cost
- Thruster Isp, locations, cant angles
- Propellant load & margins

Basic Mechanics of Propulsion...

Propulsion

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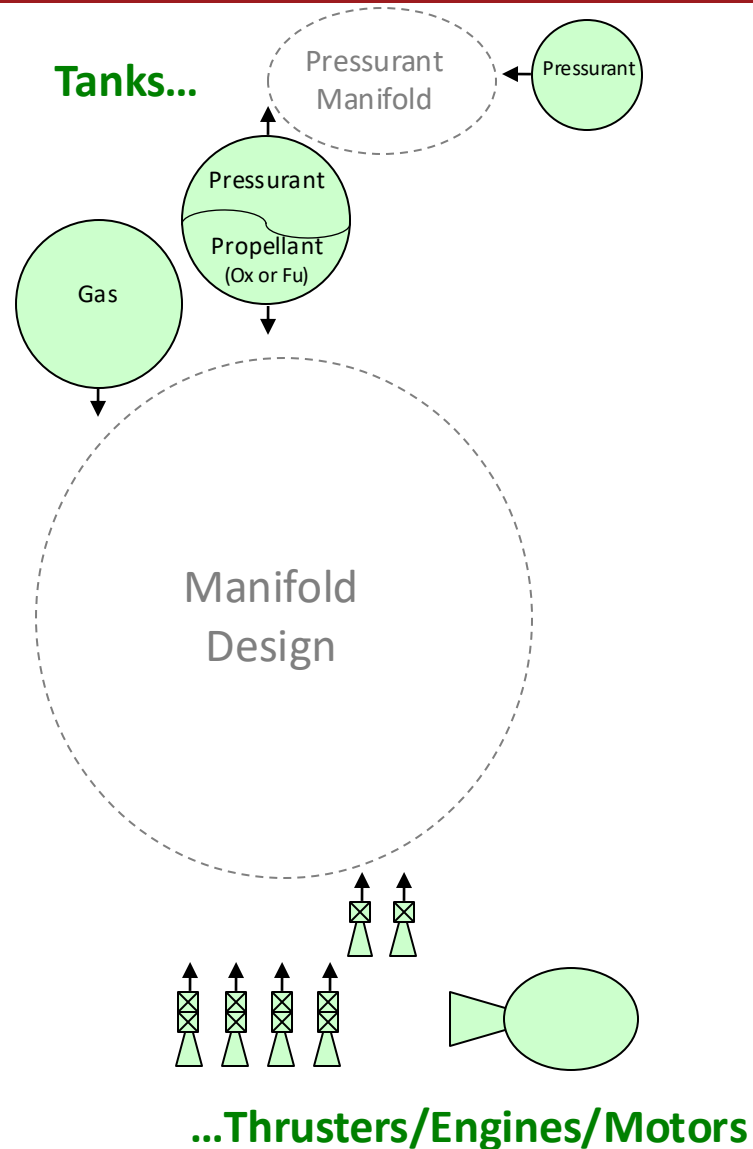
Background

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Additional Info



- **Architecture**
- **Propellant & Pressurant Tanks**
- **Thrusters**
- **Manifold**
 - Lines, filters, regulators, and flow control devices
 - Valves (solenoid, check, pyro, etc.)
 - Pressure transducers & temperature sensors
- **Redundancy and sizing**
- **Iterate...**



Common Propulsion Systems

Propulsion

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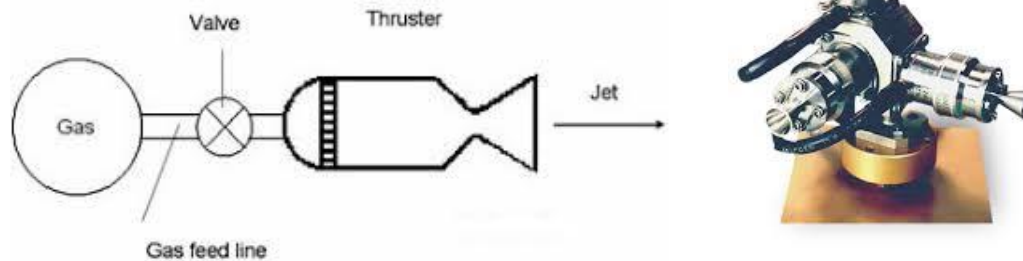
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Additional Info

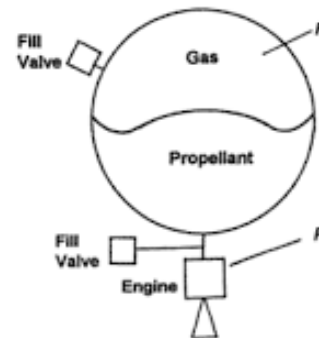
• Cold Gas Thruster Systems

- Use the expansion of a pressurized gas to generate thrust
- Typically used for orbit maintenance and/or attitude control
- Simpler system with lower mass & cost
- Lower specific impulse (45-73)



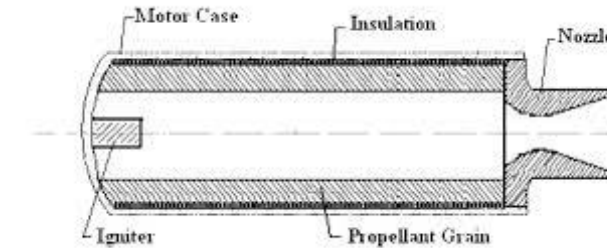
• Liquid Propulsion Systems

- Use the liquid propellant(s) to generate thrust
- Used for orbit insertion, maintenance, & attitude control
- Specific impulse varies with system type
 - Monopropellant (200-235)
 - Bipropellant (274-467)
 - Dual-mode (varies, 200-467)
- Complexity and cost vary
 - Traded with efficiency and mass



• Solid Rocket Motors

- Use the combustion of a solid propellant to generate thrust
- Typically used for ascent & orbit insertion
- Simpler design (no moving parts)
- High specific impulse (290-304)



• Electric Propulsion

- Uses electrical power to accelerate a propellant (via an electrical or magnetic field) to generate thrust
- Typically used for interplanetary trajectories, orbit maintenance, & attitude control
- Increased complexity & cost
- Very high specific impulse (500-3,000)



OSIRIS-REx: Asteroid Sample Return Mission

(Currently, on its way back to Earth.)

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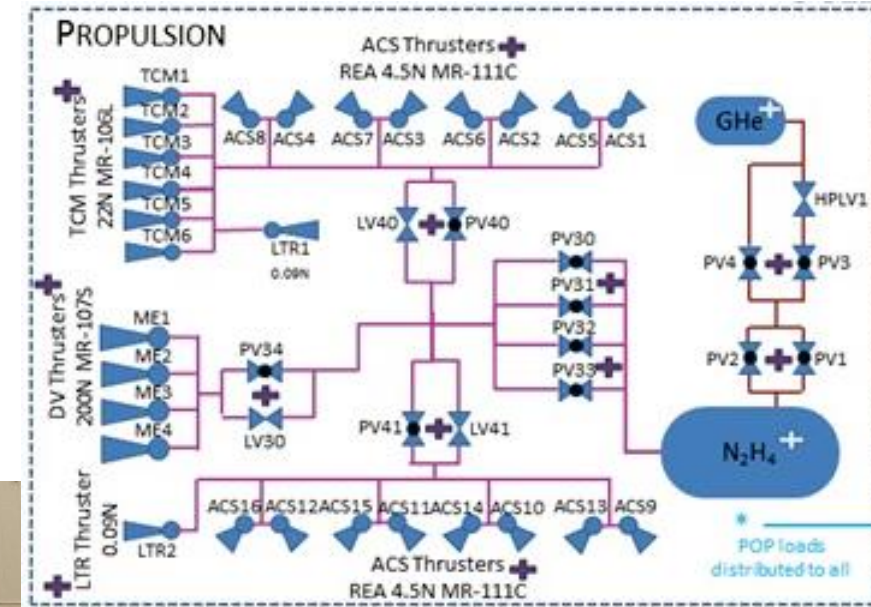
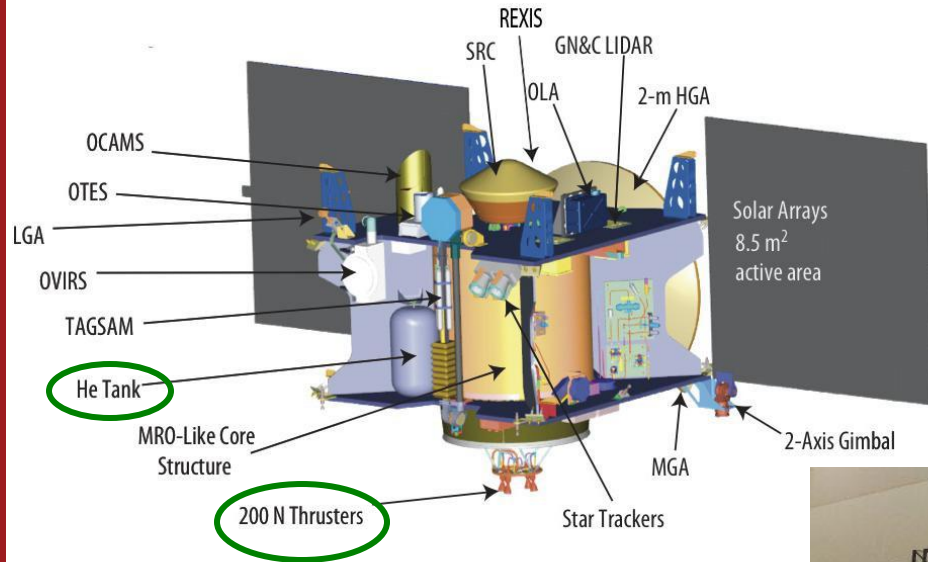
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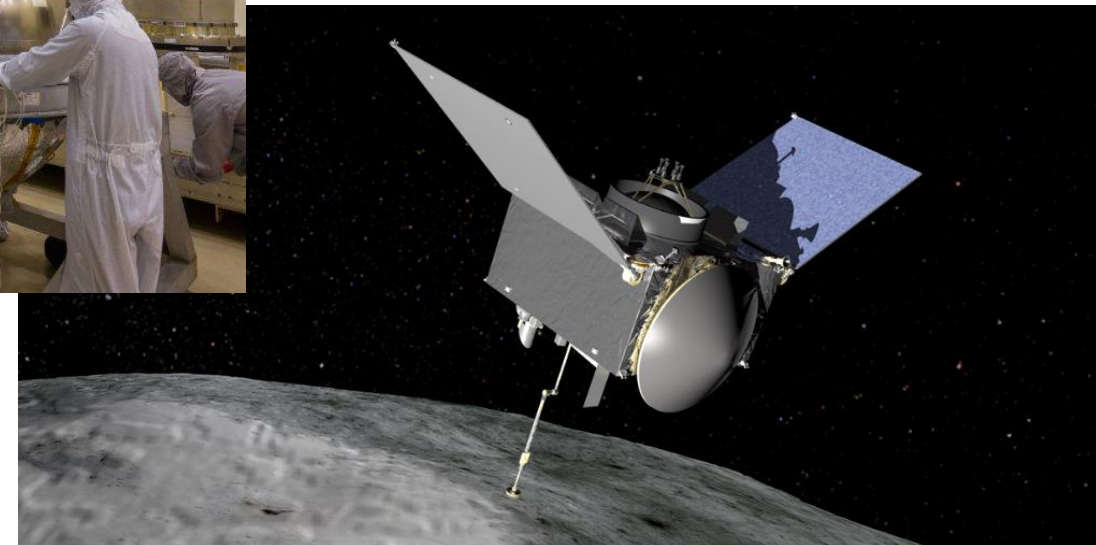
Integrated System

Additional Info



OSIRIS-REx has a monoprop system with a total of 28 engines:

- 4 high-thrust main engines
- 6 medium-thrust engines
- 16 attitude control thrusters (2/corner)
- 2 specialized low-thrust engines for the Touch-and-Go sample collect



Cassini: Saturn Orbiter

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- **Examples**

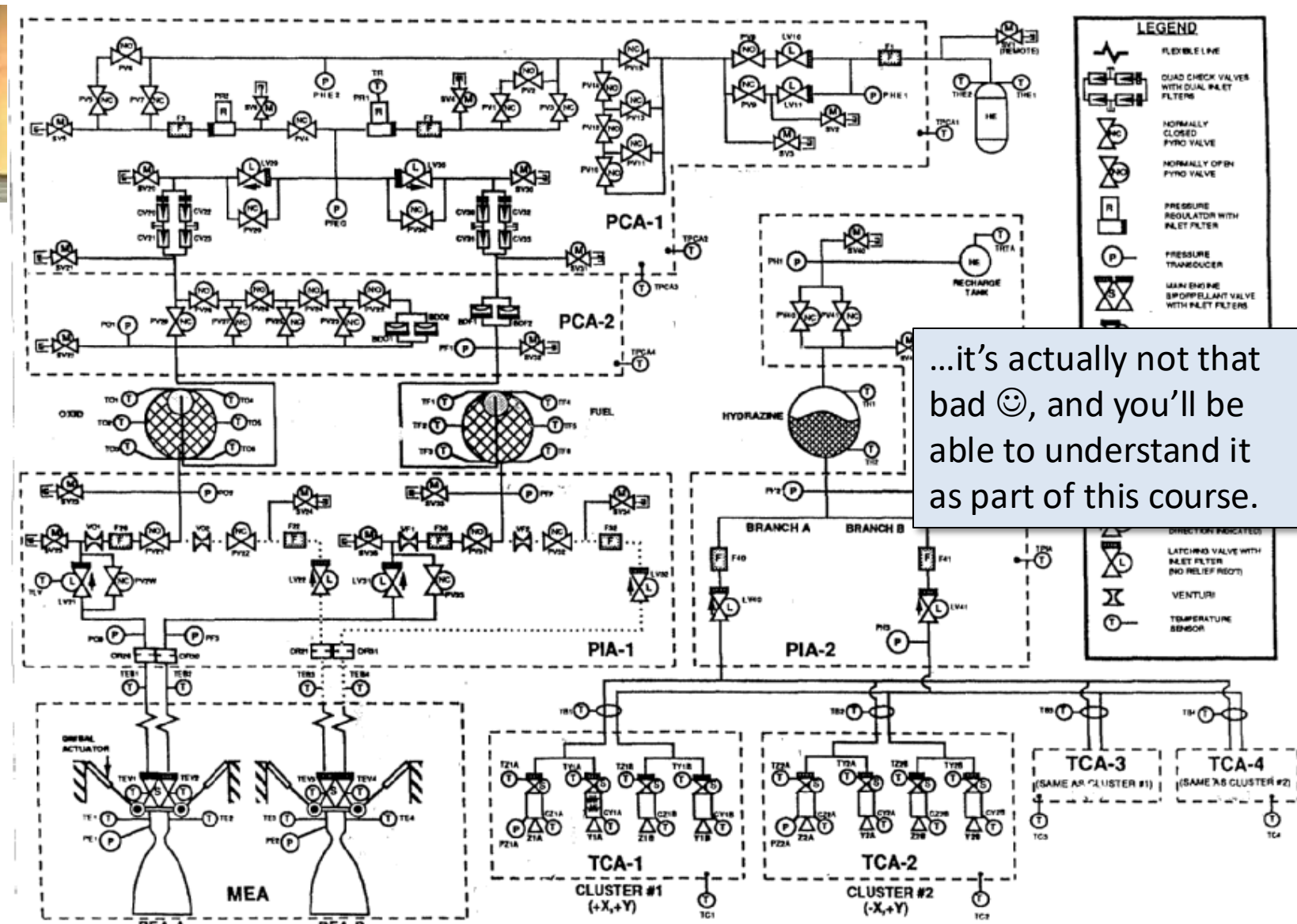
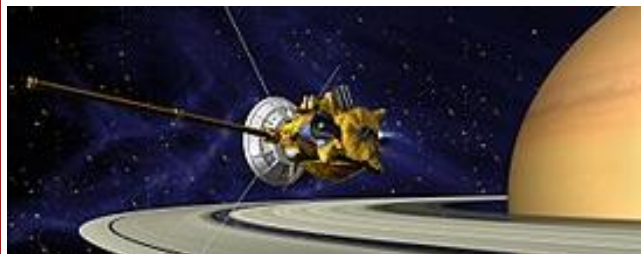
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Rocket Equation

Aka, Tsiolkovsky rocket equation
Classical rocket equation
Ideal rocket equation

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- **The Rocket Equation is used to calculate the propellant required to generate changes in velocity (ΔV)**
- **It applies to the acceleration of vehicles that produce thrust by expelling mass**
 - It is based on the conservation of momentum
 - Momentum (P_1) at T_1 = Momentum (P_2) at T_2
 - Momentum = mass x velocity; $P = mv$
 - Using this principle, the rocket equation can be derived as:

$$m_0 = m_f e^{\Delta V / v_e} \quad \text{Or, equivalently...} \quad m_f = m_0 e^{-\Delta V / v_e}$$

- Where m_0 = initial mass, m_f = final mass, ΔV = change in velocity, and v_e = exhaust velocity

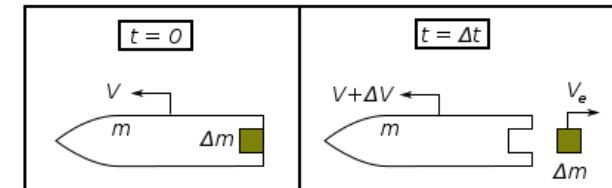
- **With thrusters, we typically characterize their efficiency using specific impulse (or I_{sp}), where:**
 - Exhaust velocity (v_e) = $g_0 I_{sp}$

- **Relating back to spacecraft, we have...**
 - Initial mass (m_0) = spacecraft wet mass
 - Final mass (m_f) = spacecraft dry mass
 - Propellant mass (m_p) = $m_0 - m_f$
 - This assumes a total ΔV and propellant, but this can also be applied sequentially to individual burns, which is necessary if the dry mass changes (eg, a probe is released)

Related concepts of dynamics

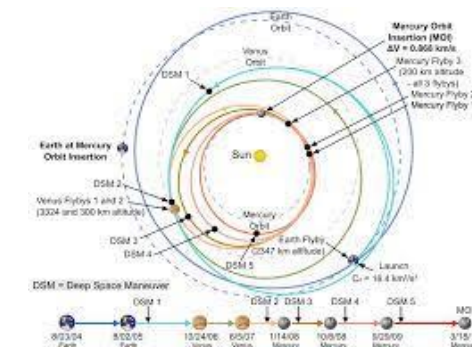
	I	II
1st law	inertia m	momentum $p = mv$
2nd law	force law $F = ma$	impulse-momentum theorem $J = \Delta p$
3rd law	action-reaction $+F_1 = -F_2$	conservation of momentum $\Sigma p = \Sigma p_0$

From www.physics.info/momentum-conservation



From Wikipedia, Rocket Equation

Messenger Trajectory (many ΔV events)



Other Common Use Cases

- Launch & landings
- Inclination changes
- Orbit maintenance
- Attitude control
- Momentum dumps

The Ideal Gas Law (aka, general gas equation)

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- In propulsion systems, gases are commonly used either directly as a means of generating thrust (eg, cold gas systems) or indirectly to pressurize the liquid propellant.
- To function within specifications, thrusters require gas/liquid propellant to enter the thruster at a specified pressure.
- The ideal gas law is used to determine how much pressurant & volume is required.
- The equation provides the relationship between pressure, temperature, and volume of a hypothetical ideal gas, which is a useful approximation for the majority of gases used in propulsion systems.
- The equation is: $PV = nRT$

Where: P = pressure (pascal)

V = volume (m³)

T = temperature (K)

n = amount (kg)

R_s = specific constant (J / kg K)

Or, frequently, we'll use bar (pressure) and liters (volume), and then the equation $PV = nRT / 100$

- In some instances (eg, hydrogen gas), other equations/factors need to be used.

Pressurized Liquid Propellant System

From faa.gov, Rockets & Launch Vehicles

Cold Gas System

From faa.gov, Rockets & Launch Vehicles

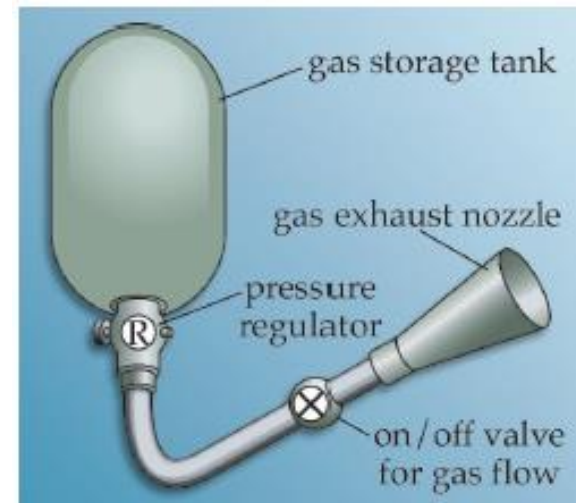


Figure 4.2.1-12. A Cold-gas Thruster. A cold-gas thruster is perhaps the simplest example of a rocket. In a typical thruster, shown in the cross-sectional drawing (upper), a gas enters from the right and stays behind the solenoid seal until it opens on command, releasing the gas through the nozzle. (Courtesy of Polyflex Aerospace, Ltd., U.K.)

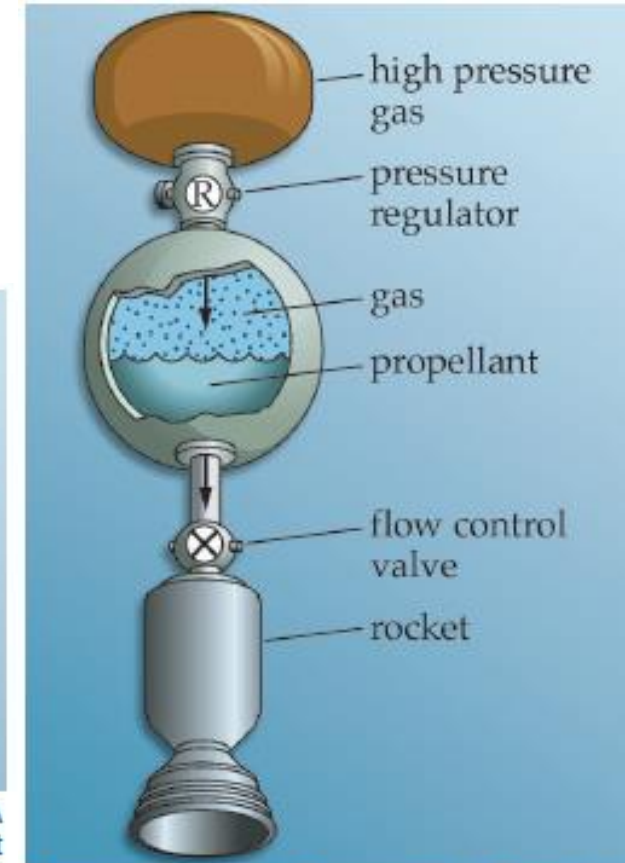


Figure 4.2.1-16. Pressure-fed Propellant System. In a pressure-fed propulsion subsystem, high pressure gas forces the liquid propellant into the combustion chamber under pressure, much like blowing liquid through a

Architecture Selection: Lunar Sample Return Example

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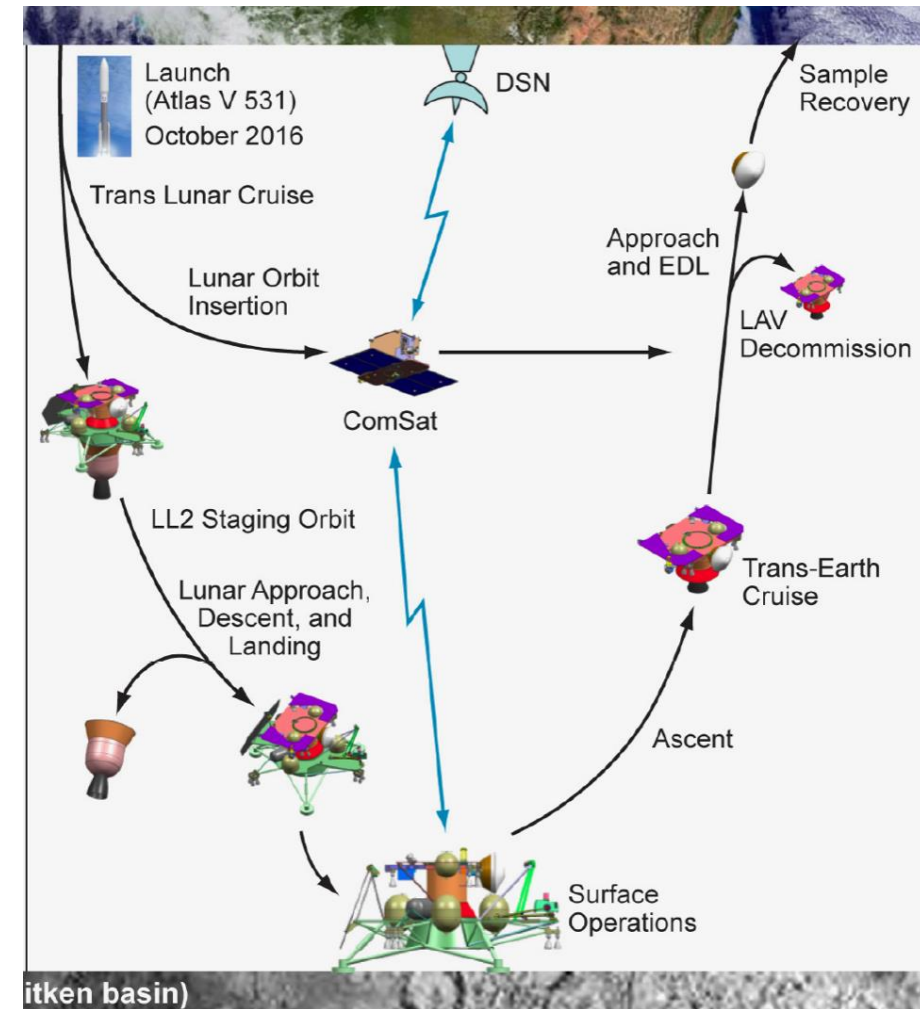
Additional Info

• Review the Concept of Operations

- Where & how much thrust is needed?
 - Trajectory maneuvers & orbit insertion
 - Orbit maintenance
 - Attitude control (propulsion versus GN&C)
 - Special considerations (eg, lunar ascent)
- What type of system accommodates each use?
 - Mass efficiency & cost
 - Consolidation of different types of system
- Identify key requirements & selection discriminators
 - Delta-V and dry mass
 - Redundancy, sample contamination, optical instruments

• Perform analysis and iterate

- Consider design at both the system and subsystem level
- Develop propulsion schematic for each required propulsion system
- Select off-the-shelf tanks & thrusters per sizing analyses



*Excluding the LV, how many thrusters/engines above?
Assume each system is single-string.*

Architecture Selection: Lunar Sample Return Example

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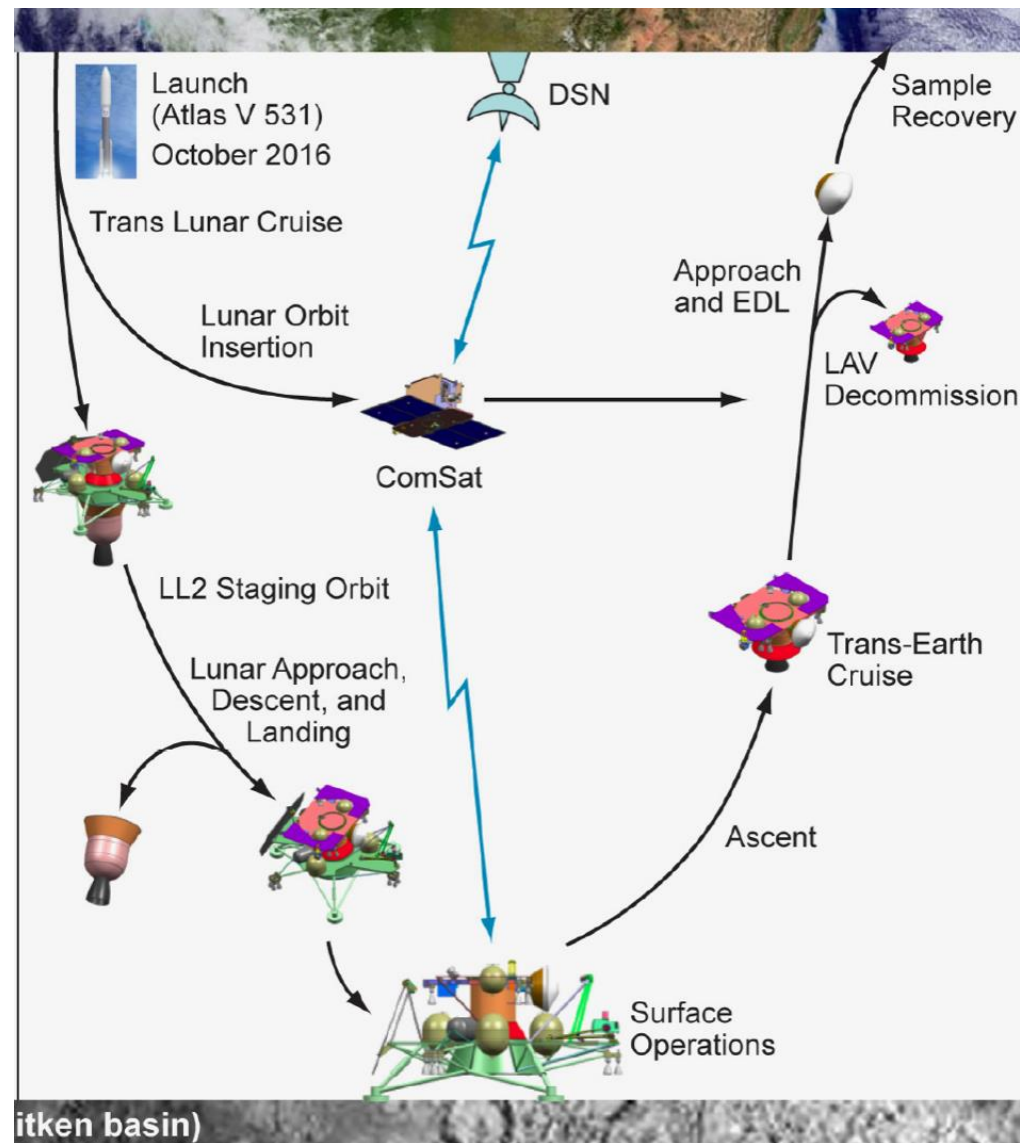
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Additional Info



• Launch

- Atlas V class Launch vehicle
- Liquid engines + SRM boosters

• Lunar Breaking & Landing System

- SRM (ejected) + Monoprop
- 12 high thrust engines

• Ascent System

- SRM + Monoprop System
- 6 high thrust engines
- 8 attitude control thrusters

• Sample Return Capsule

- None (Earth atmosphere for breaking, low mass/drag ratio for slow velocity, helicopter catch)

• Comsat

- Single-string Monoprop system
- 1 mid-thrust engine
- 8 attitude control thrusters

Total = 37 (2 SRMs + 35 thrusters, excl. LV)

Propulsion System Architecture

Propulsion

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• Requirements

• Schematic

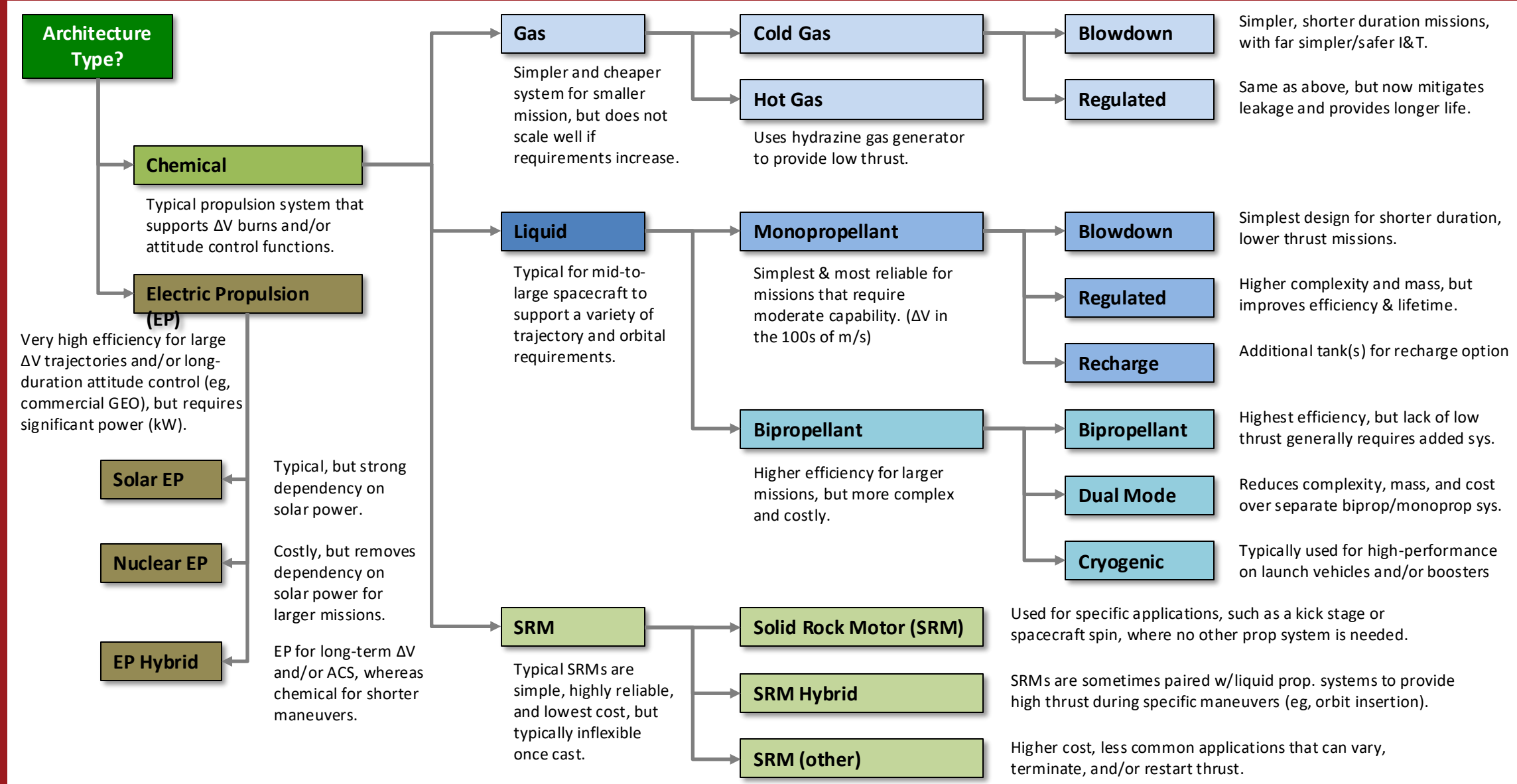
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Requirements: Example L3 & L4

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Flight System Requirements

- **L3_FS_001:** The flight system shall support the following spacecraft dry mass allocations. Note that these allocations include the propulsion subsystem dry mass.

	S/C-A	S/C-B
Dry Mass Allocation	655.8 kg	653.2 kg

- **L3_FS_002:** The flight system shall provide fully redundant 3-axis attitude and translational control to support science operations.
- **L3_FS_003:** The flight system shall provide full redundancy to ensure operations over a five-year lifetime. Note that components that are not required for the 5-year mission duration do not need to be redundant.

GN&C Subsystem Requirements

- **L4_GNC_001:** The GN&C subsystem shall accommodate a Mars gravity gradient with an average spacecraft torque of 0.00012 Nm.
- **L4_GNC_002:** The GN&C subsystem shall provide daily momentum dumps while at Mars in order to desaturate the GN&C system. Note that a reasonable thruster burn time is 5-20 sec to support a momentum dump.

Requirements: Example L4

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Propulsion Subsystem Requirements

- L4_Prop_001:** The propulsion subsystem shall support the ΔV budget specified in the table below.

Dates*	Maneuver	S/C-A ΔV (m/s)	S/C-B ΔV (m/s)
10/9/2025	Launch	-	-
10/16/2025	Phasing*	55.6	0.0
During Cruise	TCM-1	0.0	25.0
	TCM-2	3.0	3.0
	TCM-3	1.0	1.0
	TCM-4	1.0	1.0
	TCM-5	5.0	5.0
7/4/2026	Mars Orbiter Insertion (MOI)	814.5	813.7
	Gravity Loss	49.1	57.5
7/11/2026	Lower Apogee – 1	537.2	652.2
	Gravity Loss	9.2	18.4
7/18/2026	Lower Apogee – 2	537.2	652.2
	Gravity Loss	6.5	12.2
	Reserves	50.0	50.0
Total		2069.3	2291.0

* Note that a phasing event will delay spacecraft A orbit insertion by five days.

- L4_Prop_002:** The propulsion subsystem shall accommodate the following propellant masses that are separate from the propellant required for the trajectory.

	S/C-A	S/C-B
Propellant Residuals & Hold-up	24.0 kg	24.0 kg
RCS Propellant	20.0 kg	20.0 kg

- L4_Prop_003:** The propulsion subsystem shall provide thrust vector control for main engine burns via four 22-N monoprop thrusters that offset the Mars gravity losses. NOTE: You can neglect this requirement in your propellant calculations, but you should still include its contribution to your overall mass and hardware list.

Requirements: Detailed Propellant Budget

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Below is an example of a more detailed propellant budget

- The rocket equation is used to determine the propellant based on the ΔV and initial mass
- For each maneuver, a propulsion system is selected (which determines Isp)
- Additionally, there is a column for released payloads (such as a probe), which will reduce the mass/propellant for later maneuvers
- Not shown here, but sometimes included is an engine 'cant' (or incidence angle that results in the slight loss of thrust). This can be fixed or (with gimbals) articulated.

Selection of monoprop vs. biprop based on:
(1) Thrust required / burn-time, (2) efficiency, and
(3) timing relative to re-pressurization events.

Initial Mass
(eg, LV Capability)

$M_o = \text{Previous } M_f - \Delta \text{Dry}$

$M_p = M_o - M_f$

Rocket Equation
 $M_f = f(M_o, \Delta V, I_{sp})$

Delta-V Budget	Delta-V	Engine	Isp	Initial Spacecraft Mass (M_o)	Propellant Required for Maneuver	Delivered Mass (M_f)	Release of Additional Payloads
Maneuvers	m/s		s	kg	kg	kg	kg
Launch correction	30.0	Biprop	325	3994.3	37.4	3956.8	
TCM-1	50.0	Biprop	325	3956.8	61.6	3895.2	
TCM-2	215.0	Biprop	325	3895.2	254.1	3641.1	
TCM-3	20.0	Monoprop	224	3641.1	33.0	3608.1	
Jupiter Orbiter Insertion	1050.0	Biprop	325	3608.1	1012.8	2595.3	
Clean-up	20.0	Biprop	325	2595.3	16.2	2579.1	
TCM-4	70.0	Biprop	325	2579.1	56.0	2523.0	
Tour TCMs	150.0	Biprop	325	2523.0	116.0	2407.0	
TCM-5	145.0	Biprop	325	2407.0	107.1	2300.0	
Europa Orbit Insertion	665.0	Biprop	325	2300.0	433.2	1866.8	
Reduce Orbit	40.0	Biprop	325	1866.8	23.3	1843.5	
Orbital Maintenance	33.3	Monoprop	224	1843.5	27.8	1815.7	
Uncertainties in Bi-Prop	50.0	Biprop	325	1815.7	28.3	1787.5	
Uncertainties in monoprop	50.0	Monoprop	224	1787.5	40.2	1747.2	
	-		-	-	-	-	
	-		-	-	-	-	
End	End	3	4	5	6	7	8

This pattern repeats across each of the sequential maneuvers, until the final delivered mass (M_f) is the actual s/c dry mass that is available.

Schematic Diagrams

Propulsion

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- **Schematic**

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Additional Info

- **The first critical step of designing a propulsion subsystem is creating the draft schematic diagram.**
 - This diagram will help you understand the system requirements and features
- **Before you start, you need to understand the general architecture**
 - Context of overall mission & requirements
 - Type of propulsion system?
 - Cold gas, monoprop, biprop, dual-mode, SRM, electric
 - Blowdown versus regulated
 - Number and size of propellant and/or pressurant tanks
 - Number and size of thrusters
 - Type of redundancy?
 - Single-string (any single fault will cause the design to fail)
 - Selected redundancy (more critical or less reliable parts are redundant)
 - Full redundancy (two faults required for failure, although there are some common exceptions depending on the mission, such as tanks, main engine, some propellant lines, etc.)
 - If some of this information is unavailable, be sure to state your assumptions before continuing...

Approach

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• **Schematic**

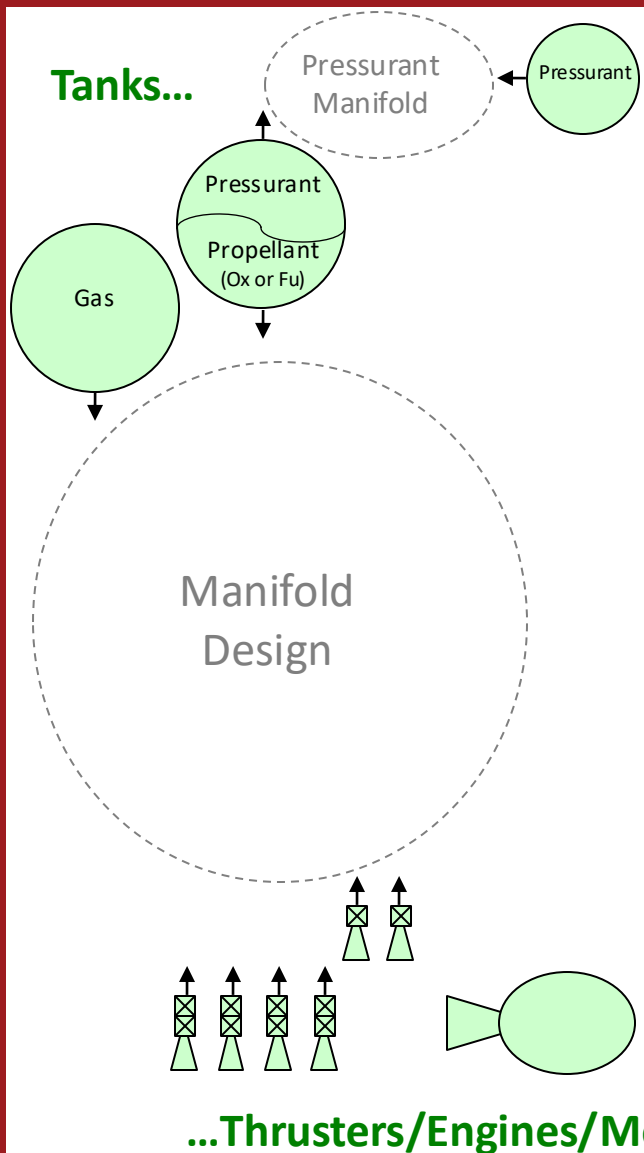
• Sizing

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Objectives

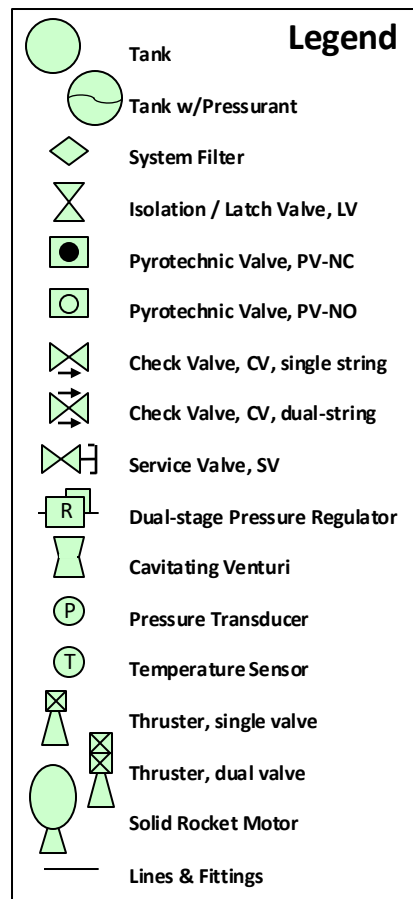
- Provide the propellant needed by the thrusters
 - Where, when, how much, etc.
- Ensure no leaks, mixing, backflow, contaminants, etc.
- Provide adequate monitoring, servicing, & testing

Approach

- Place tanks & thrusters based on architecture
- Add lines, filters, regulators, etc.
- Select and add valves (select types based on function)
- Add pressure & temperature monitoring

Review & Iteration

- Review for required redundancy
- Iterate at system & subsystem levels to minimize cost for the appropriate level of risk



Consider what you need, and add one component at a time. There is rarely a perfect answer and individual designs will vary on a variety of constraints and acceptable level of risk. Be able to defend your selections.

Component Symbols and Descriptions

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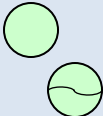
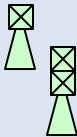
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	Tank (w/ & without pressurant)	Used to store propellant and include (for liquid systems) internal devices for separation of propellant and pressurant, along with reducing propellant movement (slosh).		Dual-stage Pressure Regulator	Controls downstream pressure (within certain constraints) mechanically or electrically. Dual stage ensures steady flow rate.
	System Filter	Removes unexpected particulates from the fluid. Typically placed just downstream of the tank(s) and service valves.		Cavitating Venturi	Provides control of pressure drops independent on downstream fluctuations. Useful in maintaining oxidizer/fuel mixture ratio and/or preventing water hammer.
	Isolation Valve / Latch Valve (LV)	Provide controlled open/closed flow that remains in position without the application of continuous electric power. Required between tank and thruster, but also useful for isolating thruster sides (eg, A vs. B), multiple tanks, etc. Often called solenoid valves as well.		Pressure Transducer	Monitors pressure to support both ground and flight operations (for safety, performance, & anomalies).
	Pyrotechnic Valve (PV) (normally closed vs. normally open)	Single-use valves that start in either an open or closed position, and then can be changed (via pyro activation) once. Useful for either isolating systems on ground for safety (and then opened in flight) or in flight (closed to prevent leaks early or late in long missions). NO = Normally open, NC = Normally closed		Temperature Sensor	Monitors temperature to support ground testing and flight operations. Placed on lines throughout system, but typically not shown on a schematic.
	Check Valve (CV) (single/dual redundancy)	Allows fluid to only a flow in one direction and, for example, prevent unintended fuel/oxidizer mixing. These can be single or redundant.		Thruster (single/dual valve redundancy)	A small rocket engine used to adjust the course or attitude of a spacecraft. These can have single or redundant valves representing by X's. If there are multiple types of thrusters, the size is often used to represent the relative thrust-levels.
	Service Valve (SV)	Positioned to both fill & drain tanks and functional testing of end-to-end system		Solid Rocket Motor	Solid Rocket Motor (only requires data/power interface, which is not required in schematic)
				Lines & Fittings	Typically titanium or stainless steel.

For component pictures & specs, see

- <https://www.space-propulsion.com/index.html>
- <https://www.moog.com/products/propulsion-controls/...>
- <https://www.valcor.com/missiles-and-aerospace/>

Sources

- SMAD, FSS, wikipedia, etc.

Interesting Failures

- <https://www.thespacereview.com/...>
- <https://spacenews.com/faulty-valve...>

Note: The above table represents common components and symbols, but it's not fully comprehensive and individual propulsion systems and symbols may vary.

Other Symbol Variations...

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• **Schematic**

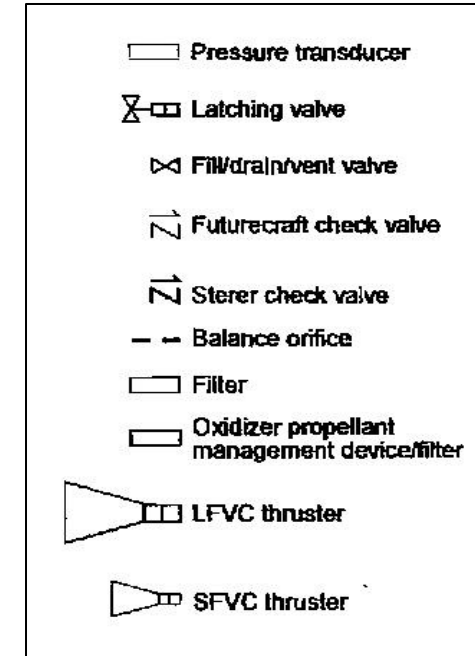
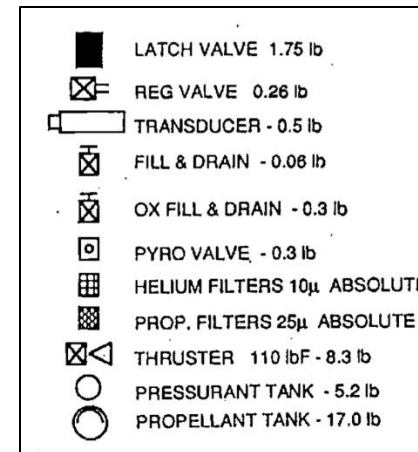
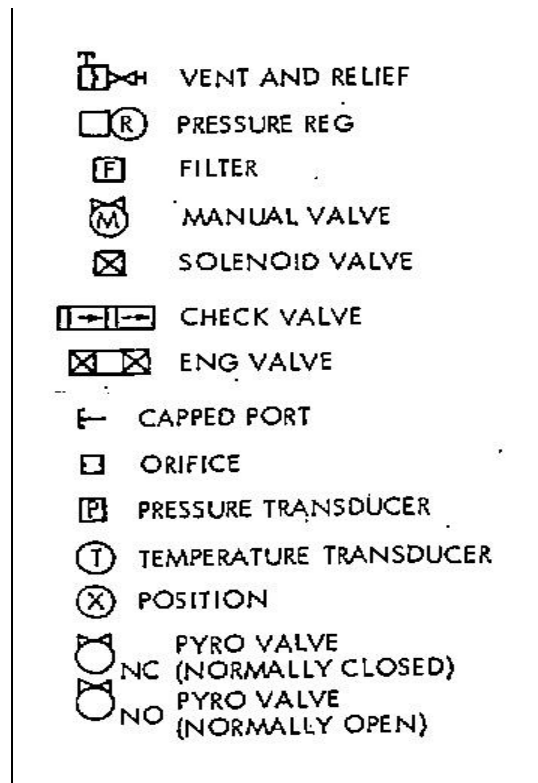
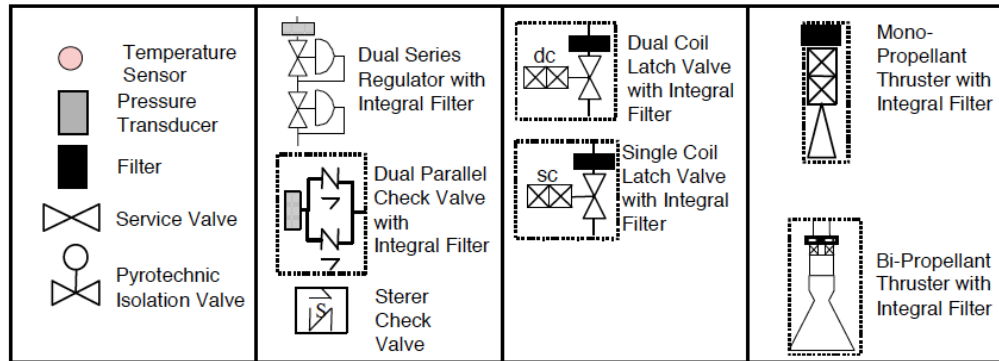
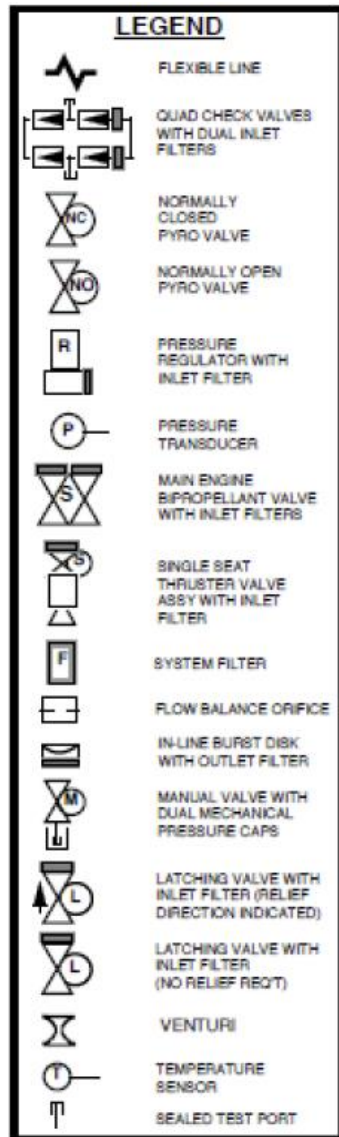
• Sizing

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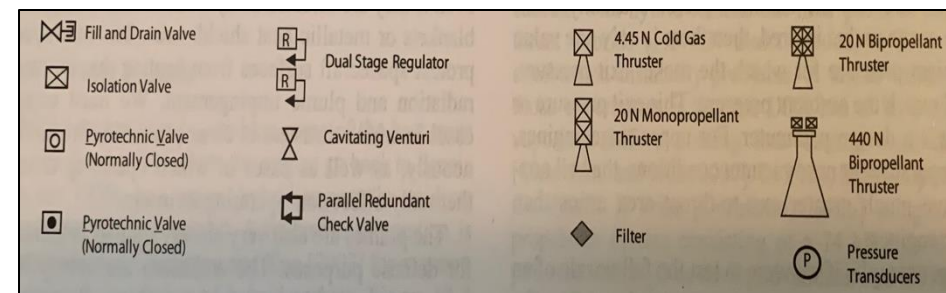
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Pulled from many existing propulsion system schematics, along with SMAD



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• **Schematic**

• Sizing

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Examples

For each example, identify each component & understand its function.

Cold Gas System

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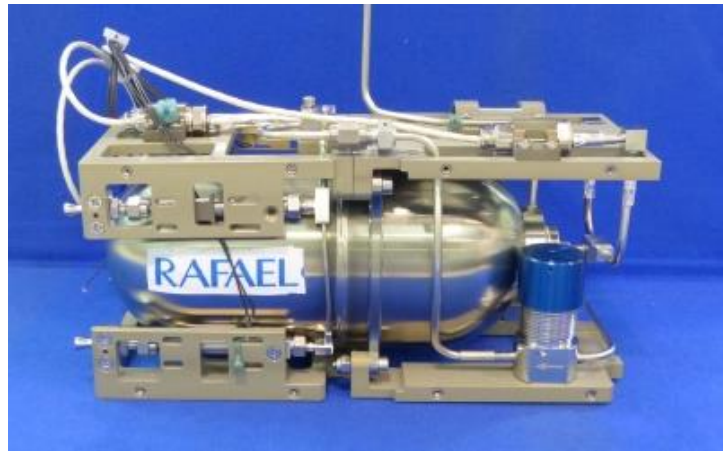
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- **Schematic**
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Hardware

Integrated System

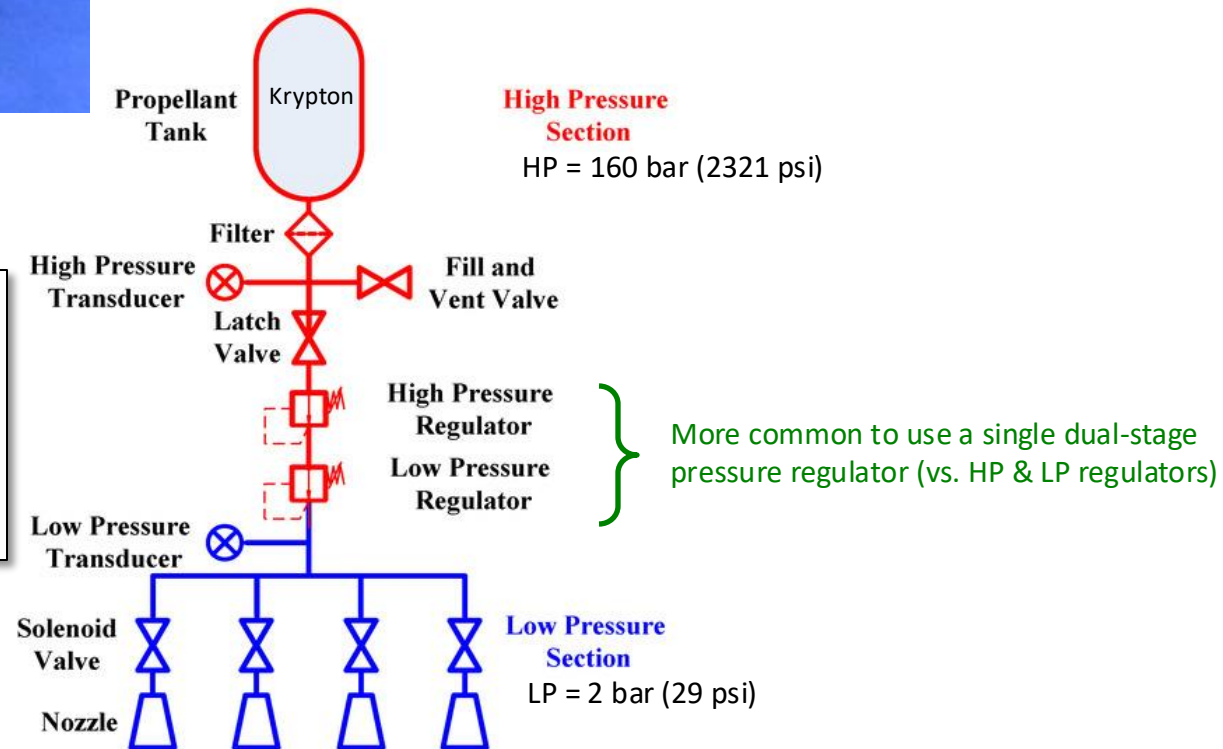
Additional Info



The propulsion system consists of the propellant tank, filter, high pressure latch valve, pressure regulators, and a four-branch manifold that feeds into four thruster assemblies. Each thruster assembly includes one solenoid valve and a nozzle through which the propellant is released. In addition to the components listed above, the propulsion system also includes a high pressure transducer, a low pressure transducer, and a fill and vent valve for propellant loading.

Propulsion System Requirements

- $\geq 150 \text{ N s}$ total impulse
- $\leq 2 \text{ kg}$ for prop system mass
- Non-toxic, non-flammable, non-explosive propellant
- 80 mN total thrust



From: "Fabrication and Testing of the Cold Gas Propulsion System Flight Unit for the Adelis-SAMSON Nano-Satellites"

Adelis-SAMSON satellite cold gas propulsion system

Simple Monoprop Blowdown Systems

Propulsion

Introduction

Background

Application

- Architecture
- Requirements

• Schematic

- Sizing
- Assignment

Hardware

Integrated System

Additional Info

• Assumptions

- Diaphragm tank with no external pressurant system
- One-year mission life

• Notes

- Additional service valve added after pyro valve to ensure adequate testing
- Since the pyro valve is more likely to add contaminants (rather than be affected by contaminants), the filter is placed after the pyro valve and prior to the latch valve.

Pressure monitor

P

F/D Valve

Hydrazine

Note that service (fill & drain valves are located at the top & bottom of the tank to aid ground operations).

F/D Valve

PV

Pyro valve enables the system for use

Filter

LV

Latch valve controls the propellant flow to the thrusters.

Thrusters

Thrusters

Monoprop Blowdown System

Propulsion

Introduction

Background

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- Architecture
- Requirements

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• Assignment

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Integrated System

Additional Info

Overview

- Global Precipitation Measurement (GPM)
- Mass = 3850 kg, power = 1.95 kW
- RCS (Reaction Control Subsystem) uses a chemical propulsion system for attitude control, reaction wheels momentum dumps and Low Earth Orbit maintenance

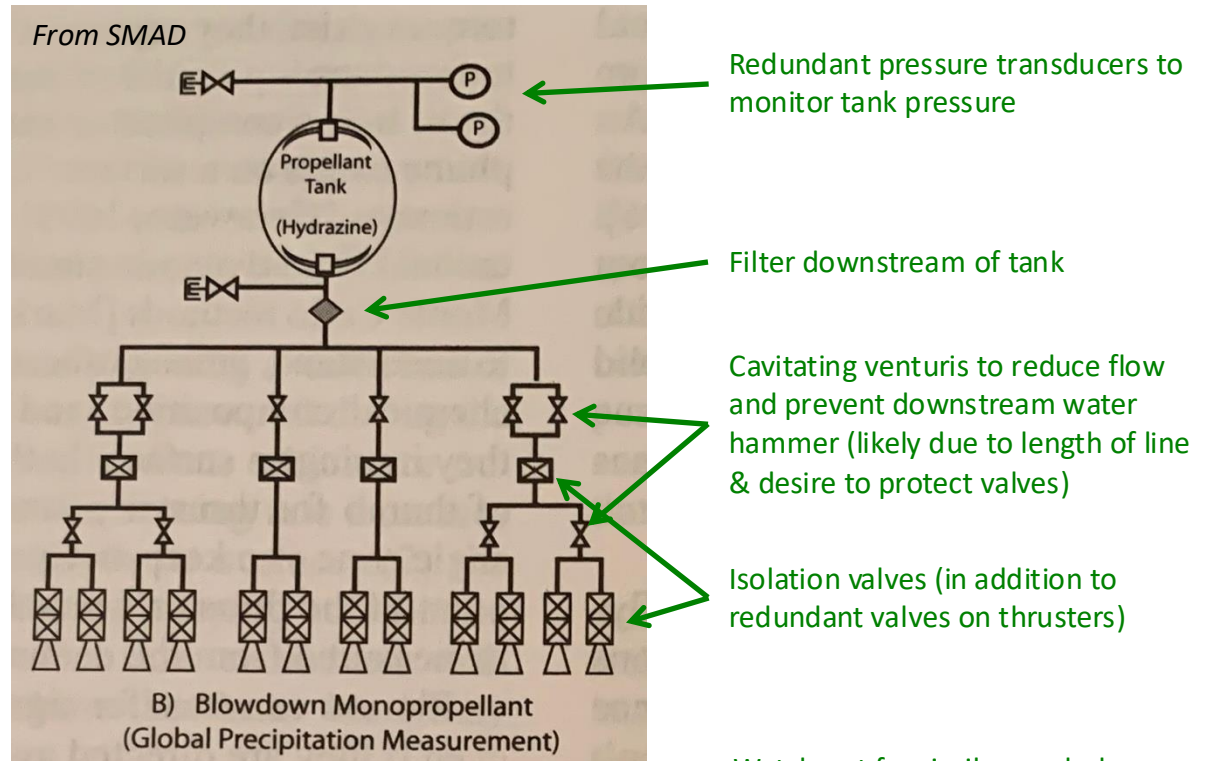
Thrusters

- 12 total thrusters
 - 8 aft (4 straight, 4 90-deg nozzles)
 - 4 forward-facing
- All thrusters are for attitude and RW dumps
- Only forward thrusters are for maneuvers
- Generate thrust by catalytic decomposition of hydrazine using heated platinum/palladium catalyst beds
- Thrusters provide
 - 44.5 N at 27.6 bar (launch)
 - 13.3 N at 6.8 bar (end of life)
- Attitude control is accomplished in pulse mode, whereas orbital maneuvers use steady state

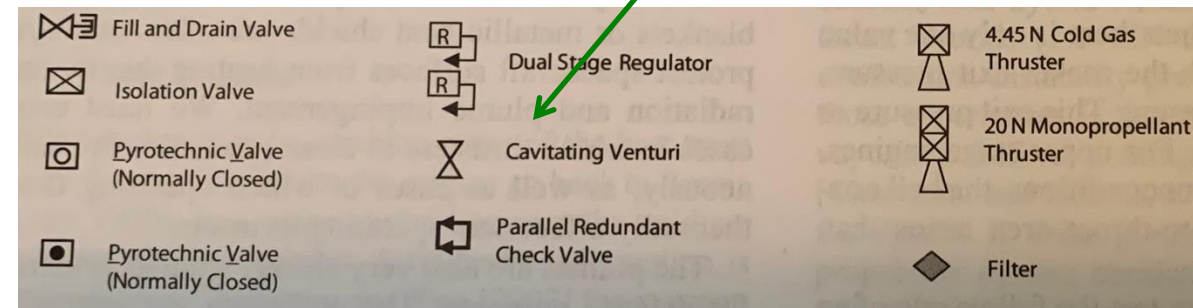
See how the thrust decreases over the mission duration due to 'blowdown' design.

Tank

- At launch, holds 545 kg hydrazine at 27.6 bar
- Qualified to 34.5 bar w/55.2 burst press.
- Minimum flight pressure is 6.8 bar.
- 2 to 50 °C temperature range with a ten-year minimum storage life of the hydrazine
- Tank pressurization is w/6.2 kg of N₂



Watch out for similar symbols with different meanings



Another Monoprop Blowdown System

Propulsion

Introduction

Background

Application

- Architecture
- Requirements

• Schematic

- Sizing
- Assignment

Hardware

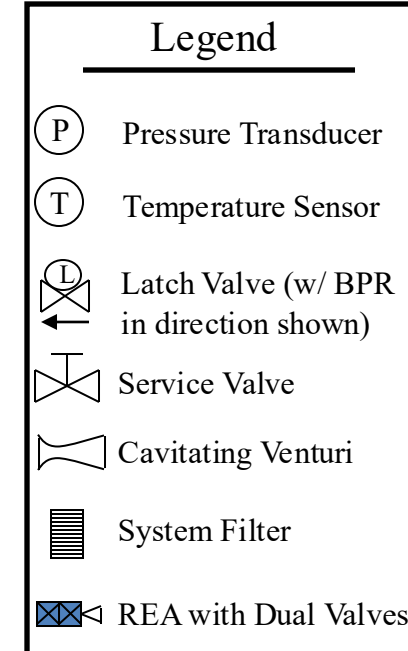
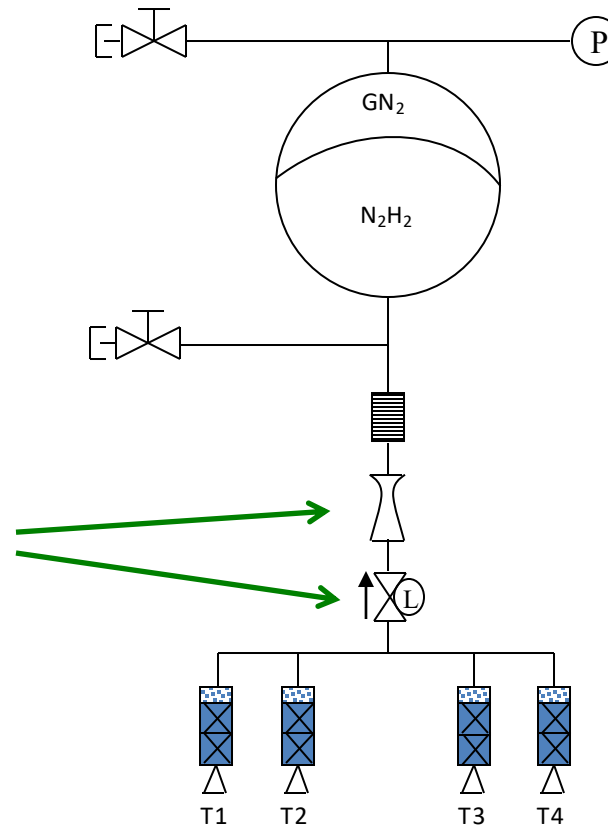
Integrated System

Additional Info

Low cost technology demonstrator (EO-1)

- EO-1 had magnetic torquers, reaction wheels, & electric propulsion, so monoprop system was primarily for orbit raises

Combines a cavitating venturi with a back pressure regulator (BPR) to maintain a defined downstream pressure



Biprop Main Engine Assembly (MEA)

Propulsion

Introduction

Background

Application

• Architecture

• Requirements

• Schematic

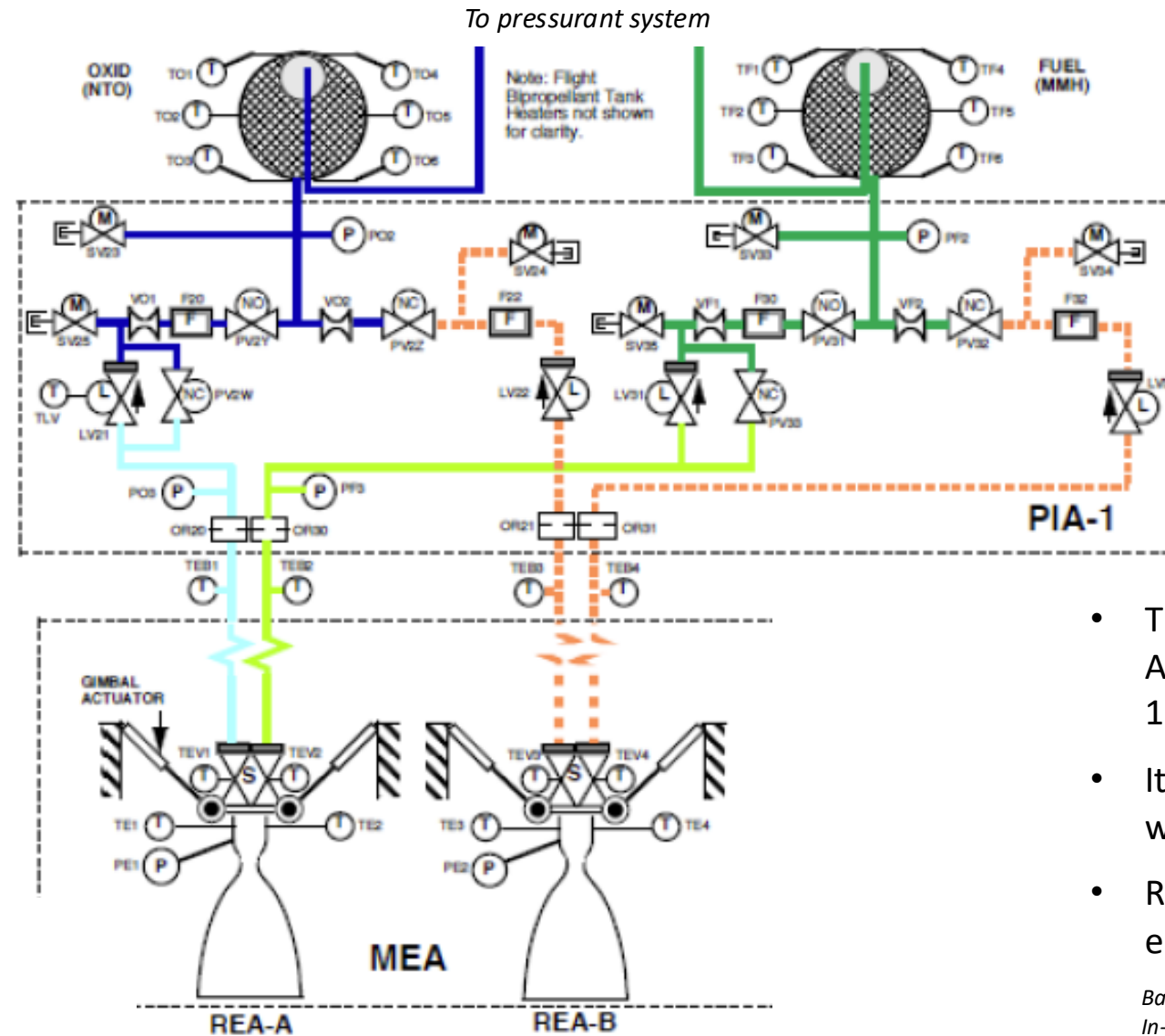
• Sizing

• Assignment

Hardware

Integrated System

Additional Info



- This is the Cassini Main Engine Assembly for orbit insertion and > 100 other maneuvers.
- It is a fully redundant biprop system with NTO & MMH
- REA-B was the redundant main engine and was ultimately never used

Barber, T. "Final Cassini Propulsion System In-Flight Characterization." AIAA, 2018.

Biprop Main Engine Assembly (MEA)

Propulsion

Introduction

Background

Application

- Architecture
- Requirements

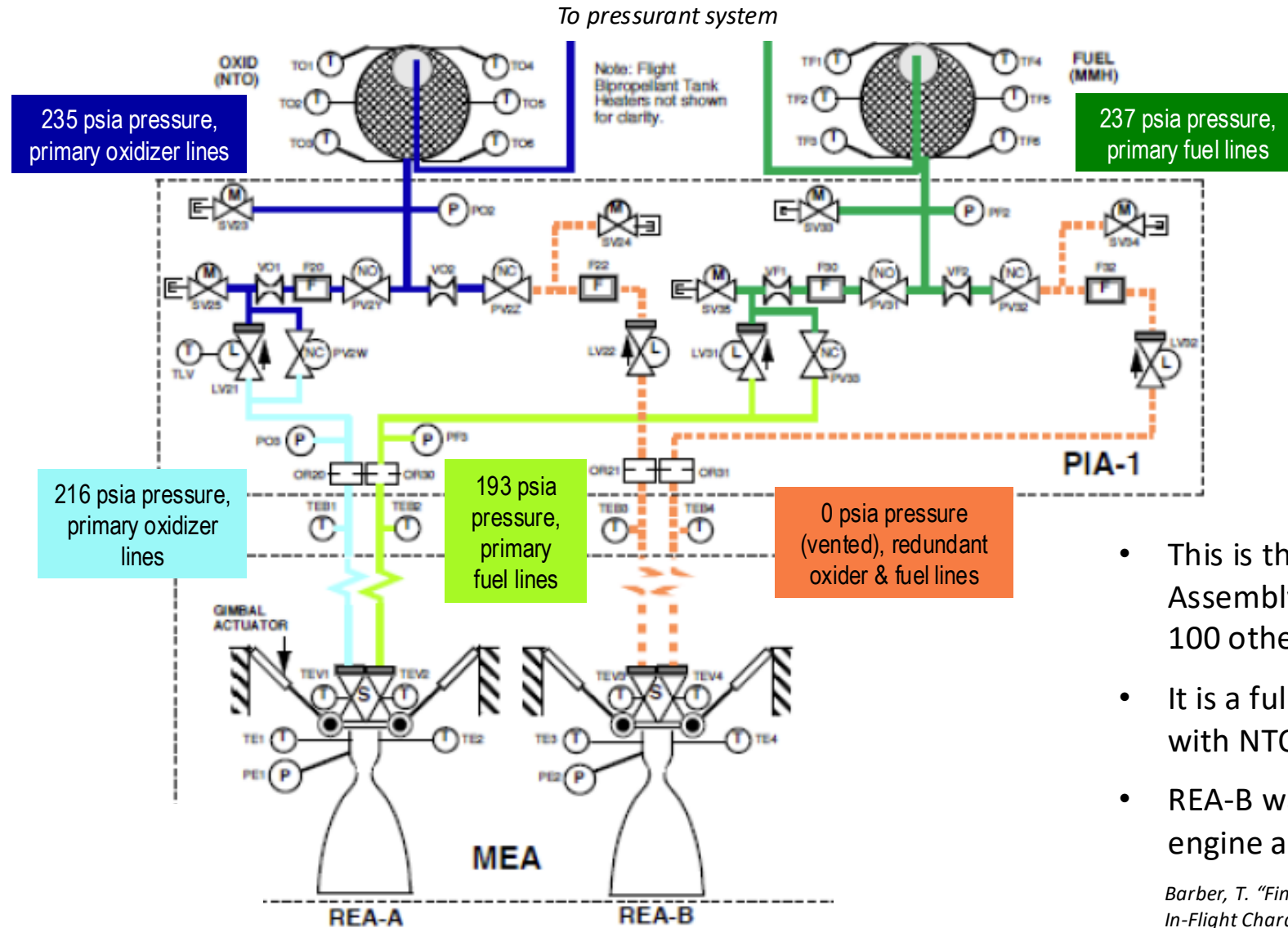
• Schematic

- Sizing
- Assignment

Hardware

Integrated System

Additional Info



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- It is a fully redundant biprop system with NTO & MMH
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Biprop Main Engine Assembly (MEA)

Propulsion

Introduction

Background

Application

- Architecture
- Requirements

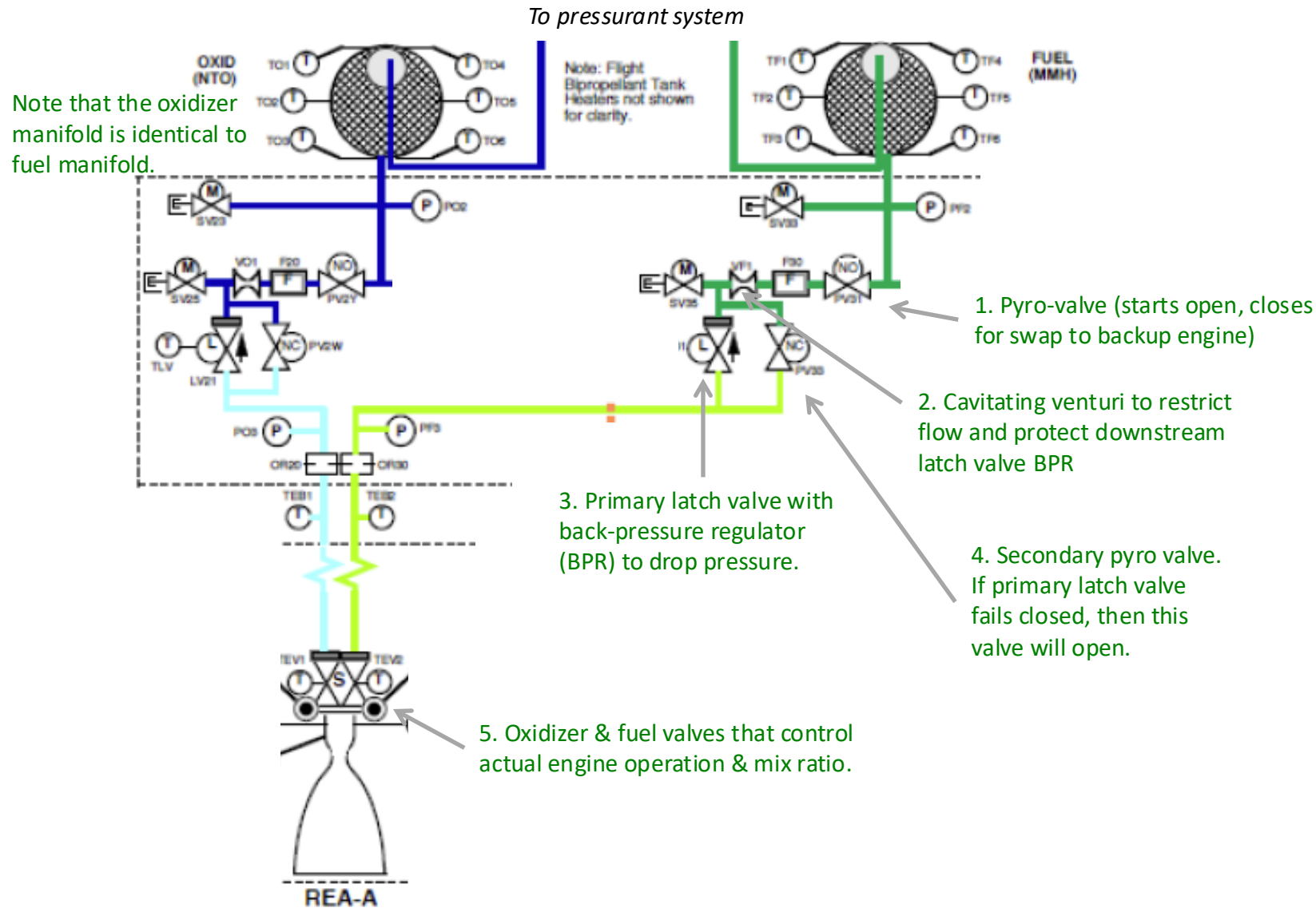
• Schematic

- Sizing
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Hardware

Integrated System

Additional Info



Biprop Main Engine Assembly (MEA)

Propulsion

Introduction

Background

Application

- Architecture
- Requirements

• Schematic

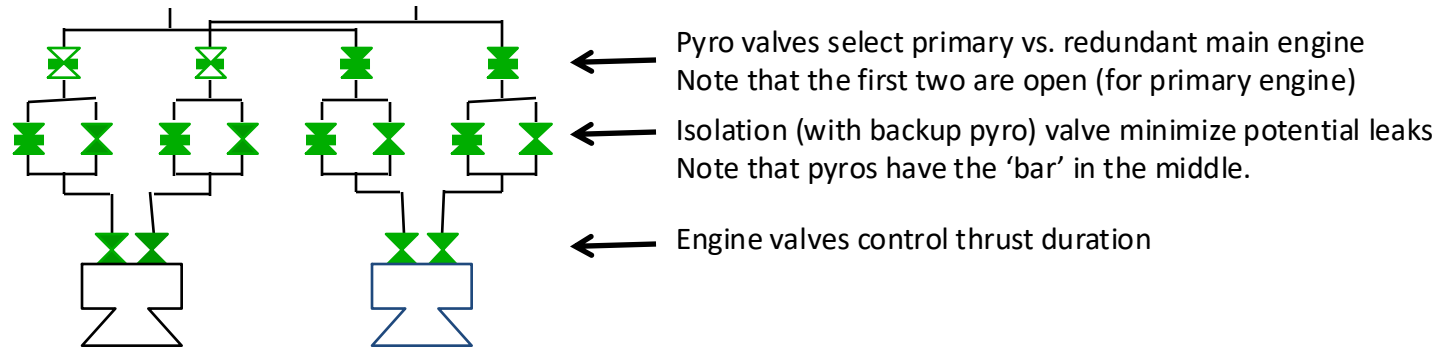
- Sizing
- Assignment

Hardware

Integrated System

Additional Info

Another view...



Biprop Main Engine Assembly (MEA)

Propulsion

Introduction

Background

Application

• Architecture

• Requirements

• Schematic

• Sizing

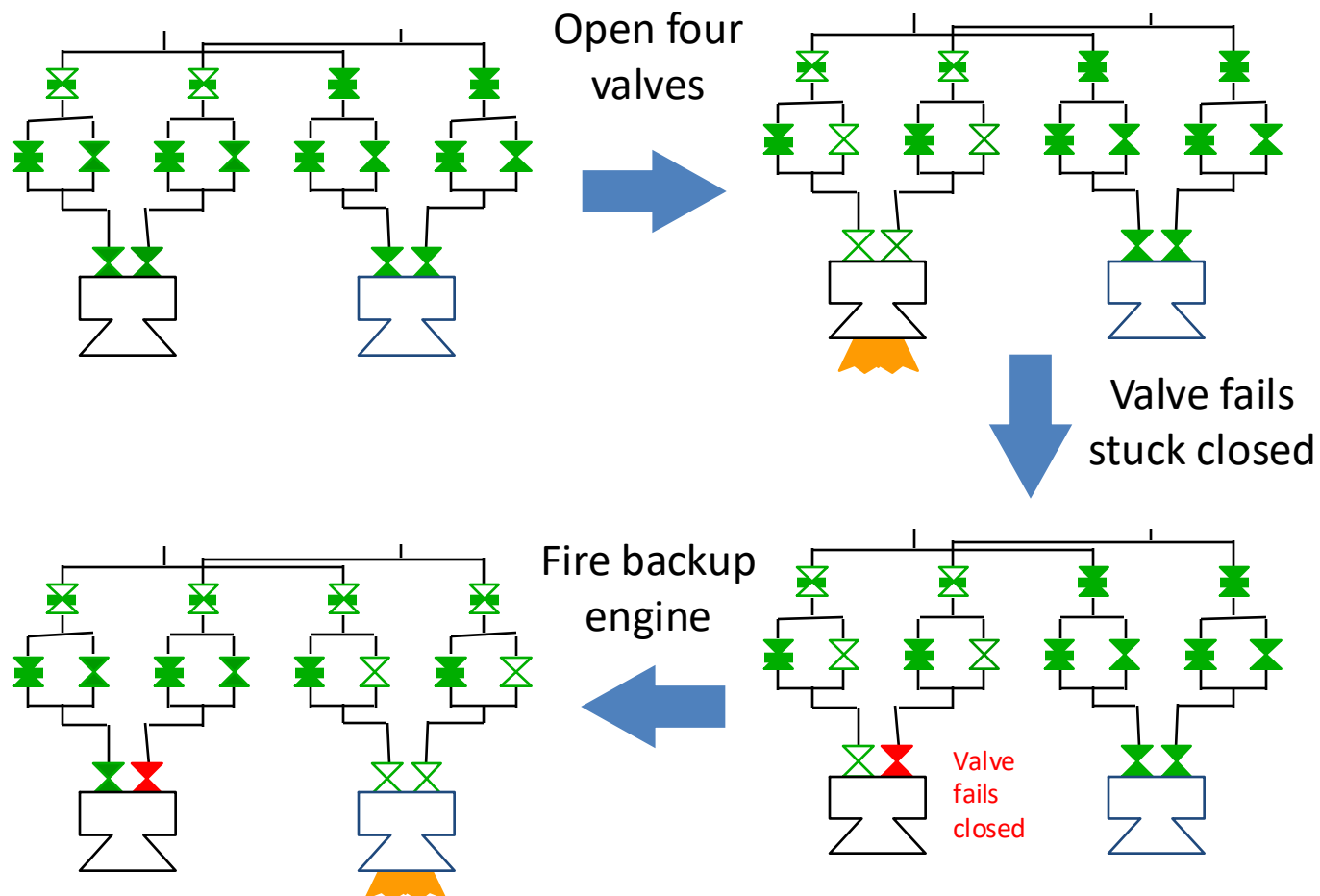
• Assignment

Hardware

Integrated System

Additional Info

If an engine valve fails...



Source: Weld, D. "CSE 592: Reconfiguring for a failed engine.", AIAA 1998

Pyro-Ladders

Propulsion

Introduction

Background

Application

- Architecture
- Requirements

• Schematic

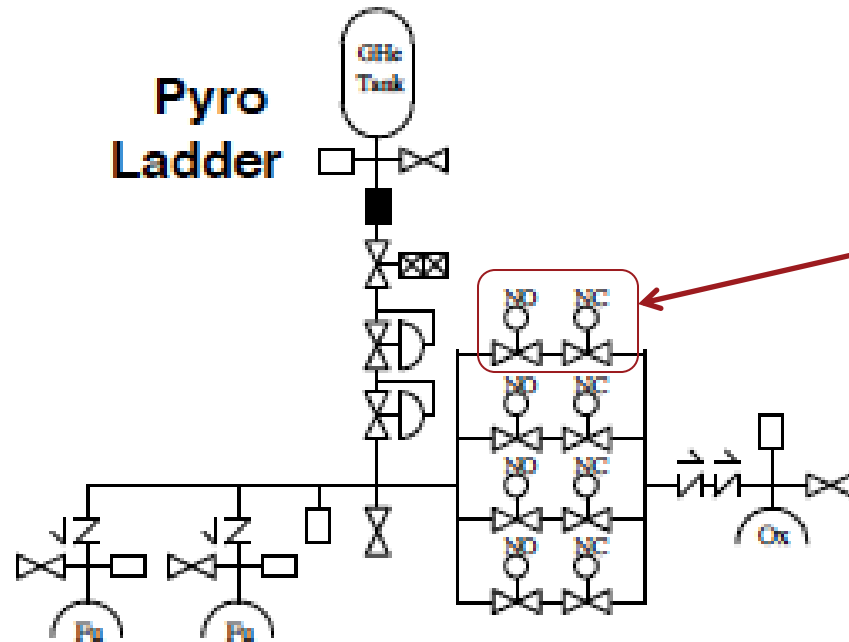
• Sizing

• Assignment

Hardware

Integrated System

Additional Info



- Pressurization using a pyro-ladder
 - Open Pyro-NC-1
 - Tank pressurizes
 - Close Pyro-NO-1
- Repeat for each “rung” of the pyro
- For example, in this schematic, the oxidizer can be pressurized 4 times

Source: Wiley, S. “Design and Development of the Messenger Propulsion System.”, AIAA 2003.

Propulsion Schematic Instructions

Propulsion

Introduction

Background

Application

- Architecture
- Requirements

• Schematic

- Sizing
- Assignment

Hardware

Integrated System

Additional Info

1. Place estimated propellant & pressurant tanks

2. Add thrusters

- Typically in groups to provide greater clarity/organization (eg, thruster types, A vs. B string, etc.)
- Use size to indicate relative thrust differences.

3. Add lines, filters, regulators, & flow control devices downstream of tanks

- Use filters to remove contaminants (eg, from pyro valves) and protect more sensitive components (eg, regulators)
- Add pressure regulators, venturis, and or back-pressure-regulators (BPRs) to protect downstream components

4. Add valves appropriate for function:

- Isolation valves for basic on/off flow control
- Pyrotechnical and/or check valves for ground or flight safety (particularly for long durations)
 - Use a “pyro ladder” that combines multiple pyro valves to produce some on/off control while minimizing leaks
- Service valves for fueling, draining, and testing
 - For example, if there are closed pyro valves, need to additional service valve to test

5. Add pressure transducers & temperature sensors for monitoring

- Pressure transducers are typically added opposite to service valves (temp sensors are usually not shown)

6. Review for desired level of redundancy

- If fully redundant, then no single fault should cause mission failure (w/exceptions of tank, SRM, main engine, etc.)

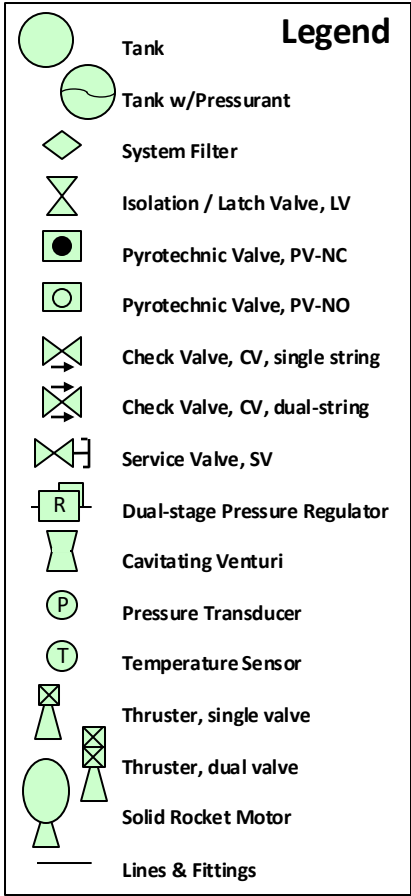
7. Additional checks & labeling

- Design to avoid back-flowing filters
- Ensure all components can be tested at the system level (ie, tested after spacecraft integration)
- Adjust sizes and add labels (oxidizer, fuel, & pressurant names, thruster & SRM names, additional notes)

8. Determine component sizing, factor in complexity/costs, & iterate

- This is done in concert with overall propulsion system design and analysis

Reminder: There is rarely a perfect answer and individual designs vary on a variety of constraints and an acceptable level of risk. This is intended to be an initial design, prior to a more detailed analysis. At this stage, the most important item is that you understand and can defend the components you’ve selected...



Propulsion Subsystem Design Steps

Propulsion

Introduction

Background

Application

- Architecture
- Requirements
- Schematic
- **Sizing**
- Assignment

Hardware

Integrated System

Additional Info

- **Step 1: Create a draft schematic**
- **Step 2: Use ΔV to determine propellant quantity**
- **Step 3: Use propellant quantity to determine tank size**
- **Step 4: Use tank size to determine pressurant quantity and tank size**
- **Step 5: Estimate wet/dry system mass**

Step 1: Determine Architecture & Draft Schematic

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• **Sizing**

• Assignment

Hardware

Integrated System

Additional Info

- **Identify Candidate Propulsion System Architecture**
- **Draft Preliminary Schematic**
 - See Propulsion Schematic class handout (charts)
 - Consider writing out the specific functions that you require
 - Number and type of tanks
 - Number and type of thrusters
 - Add pressure regulator for pressurized (rather than blow-down) system
 - Add latch valves for nominal control
 - Consider adding/swapping pyro valves/ladders to preserve pressurization and/or add redundancy
 - Add check valves & cavitating venturis to control direction and magnitude
 - Add service valves for access
 - Add pressure and temperature sensors
 - **Important**
 - For a back-of-the-envelope design (eg, exam question), you may not need to create a propulsion schematic, but you should be able to estimate the # of regulators, valves, sensors, etc.

Step 2: Calculating Propellant

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• **Sizing**

• Assignment

Hardware

Integrated System

Additional Info

- **Calculate required usable propellant, based on total ΔV , S/C mass & specific impulse**

- Use rocket equation equation to calculate required propellant mass

$$\Delta V = g_c I_s \text{LN} (M_i / M_f) \text{ (m/s)}$$

where, $g_c = 9.8067$

I_s = Ave. specific impulse (s)

M_i = Initial S/C mass (kg)

M_f = Final S/C burnout mass (kg)

- **Typically, either initial total S/C mass or final S/C burn out (dry) mass is known**
- **In each case, S/C mass includes unknown PROP mass; therefore, PROP mass must be initially estimated to yield net S/C mass**
 - $M_p = M_f [\exp (\Delta V / g_c I_s) - 1]$

Step 2: Calculating Propellant

Propulsion

Introduction

Background

Application

- Requirements

- Architecture

- Schematic

- **Sizing**

- Assignment

Hardware

Integrated System

Additional Info

- Add additional propellant to cover ACS, margin, and other needs
- Margin may be included due to:
 - Trajectory uncertainty (this is typically already included as additional ΔV)
 - Propulsion system performance variations (temperature, Isp, biprop MR, thruster, etc.)
 - Conservative estimates of thruster performance typically account for this margin
 - ACS and/or additional propellant margin
 - ACS propellant margin is typically provided by ACS subsystem (as part of their propellant estimate)
 - Often, no additional margin is carried (eg, HW2)
- Resulting total propellant is then increased to account for residuals/hold-up (ie, non-usable propellant)
 - 1% is typical for monopropellant systems
 - 3.5% is typical for bipropellant systems

Step 2: Calculating Propellant

Propulsion

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• Schematic

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Hardware

Integrated System

Additional Info

- **For bipropellant & dual mode systems, total propellant load must be split into fuel & oxidizer**
 - $M_{\text{tot prop}} = M_{\text{ox}} + M_{\text{fu}}$
and mass ratio (MR) = $M_{\text{ox}} / M_{\text{fu}}$,
therefore, $M_{\text{ox}} = M_{\text{fu}} * \text{MR}$
 - For biprop, typical assembly requires MR = 1.65
 - For $\text{N}_2\text{O}_4/\text{MMH}$, this equates to equal Fu/Ox tanks
 - Common tanks reduces the cost
 - For dual mode, typical assembly requires MR = 1.05
 - For $\text{N}_2\text{O}_4/\text{N}_2\text{H}_4$, this equates to either unequal or non-optimum tanks
 - Either a cost hit or mass penalty

Step 3: Calculate Tank Size

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• **Sizing**

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Hardware

Integrated System

Additional Info

• Number of Tanks

- The number of tanks for fuel and oxidizer (as well as pressurant) is typically based on the amount of propellant versus available heritage tanks
- Goal is to select the type & quantity of tanks that minimizes mass (& cost)
- Note that wherever possible, identical tanks should be used
 - For example, oxidizer & fuel or multiples of each
 - This is typically much cheaper versus manufacturing different tanks

• Tank Size

- Cold gas: See pressurant sizing on the following charts
- Monoprop & Biprop
 - Use propellant density to determine the required volume
 - Add initial 20% for margin against total tank capacity
 - Also, need to add the pressurant for simple monoprop blowdown (see following charts)
 - Select heritage tanks that meet the volume specification

• In concert with pressurant calculations, iterate through the options to find the minimum mass

- Note that mass is often used as a proxy for cost when cost #'s are unavailable

Step 4: Determine Pressurant...

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• **Sizing**

• Assignment

Hardware

Integrated System

Additional Info

- **The role of pressurant is to ensure adequate propellant pressures at the thruster inlet nozzle**
 - For example, a typical cold gas thruster might require 100 psia to maintain the desired thrust level (eg, in the context of the prop system, including pressure regulators, etc.)
- **Pressurant Sizing Approach**
 - The propulsion system (and specifically mass & volume of pressurant) needs to be sized such that
 - At the beginning of mission (BOM), the pressure is within the tank specification
 - At the end of mission (EOM) the pressure is still sufficient for the thrusters
- **The Ideal Gas Law**
 - For some monoprop/biprop systems (eg, w/GN2), the ideal gas law can be assumed
 - $PV = nRT$

Where: P = pressure (pascal), V = Volume (m^3)

n = mass (kg), R = Gas Constant ($J / kg\ K$)

T = Temperature (K)
 - The ideal gas law is used to trade mass, volume, & pressure
 - For typical spacecraft, T can be assumed at room temperature ($\sim 300\ K$)
 - Gas constants are fixed (such as $296.8\ J / kg\ K$ for GN2)

Step 4: Determine Pressurant...

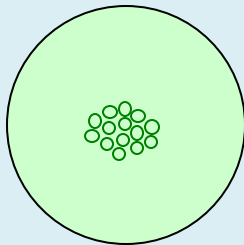
• For pressurant (gas) calculations, there are several variations, including:

- Cold Gas Propellant: Gas released is the propellant required (derived from the rocket equation)
- Monoprop Blowdown: No gas is released (constant mass), but volume increases to fill the void left by the expended liquid propellant
- Regulated Monoprop: Same as the above, but there is an additional pressurant tank (with a fixed volume) to consider

Cold Gas Propellant Approach

A) Beginning of Mission (BOM)

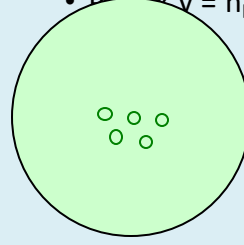
- Pressurant tank is filled to maximum expected operating pressure (MEOP)
- Using the ideal gas law, we have:
 - $P_{BOM} \times V = n_{BOM} \times 296.8 \times 300$



For example, GN2 (gaseous nitrogen) at 300 bar.

B) End of Mission (EOM)

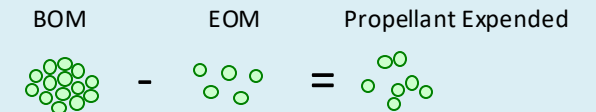
- Pressurant tank is nearly empty with the lower limit of the minimum regulator/thruster operating pressure reached
- Using the ideal gas law, we have:
 - $P_{EOM} \times V = n_{EOM} \times 296.8 \times 300$



For example, GN2 (gaseous nitrogen) at 100 bar

C) Propellant Mass

- The difference between BOM and EOM pressurants is the actual pressurant mass
- For cold gas systems, this is calculated using the rocket equation
 - $n_{BOM} - n_{EOM} = n_{propellant}$



Note that 296.8 is the specific gas constant for GN2 (J/kg K) and 300K is a standard temperature assumption.

Approach

- We can use the ideal gas equation in the following two forms
 - $P_{BOM-EOM} \times V = n_{propellant} \times 296.8 \times 300$, where $P_{BOM-EOM} = P_{BOM} - P_{EOM}$
 - $P_{EOM} \times V = n_{EOM} \times 296.8 \times 300$
- The first equation can be used to solve for volume
- And then the second equation can solve for n_{EOM}

Margin

- It's a best practice to carry margin on the tank capacity (eg, 20%) in case the ΔV requirements need to be increased. To account for this, propellant mass remains the same, but the volume that is used to calculate the residual gas pressure and mass (at EOM) should be increased by the specified margin.
- In other words, you will plan to underfill your tank slightly, but still meet your EOM pressure requirements. On the day of launch, you'll still fill to MEOP and thus convert this tank capacity margin into operations propellant margin

Step 4: Determine Pressurant...

Propulsion

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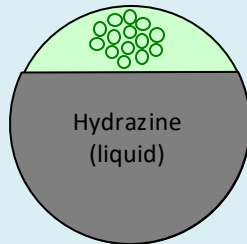
• For pressurant (gas) calculations, there are several variations, including:

- Cold Gas Propellant: Gas released is the propellant required (derived from the rocket equation)
- Monoprop Blowdown: No gas is released (constant mass), but volume increases to fill the void left by the expended liquid propellant
- Regulated Monoprop: Same as the above, but there is an additional pressurant tank (with a fixed volume) to consider

Monoprop Blowdown Approach

A) Beginning of Mission (BOM)

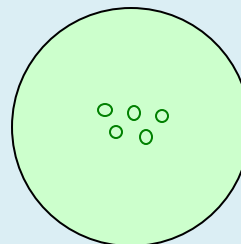
- Propellant tank holds the required liquid propellant and then is 'topped' off with pressurant (at MEOP)
- Using the ideal gas law, we have:
 - $P_{BOM} \times V_{BOM} = n \times 296.8 \times 300$



For example, GN2 (gaseous nitrogen) at 20 bar.

B) End of Mission (EOM)

- As the liquid propellant is exhausted, the pressurant fills the resulting void until the minimum pressure is reached
- Using the ideal gas law, we have:
 - $P_{EOM} \times V_{EOM} = n \times 296.8 \times 300$

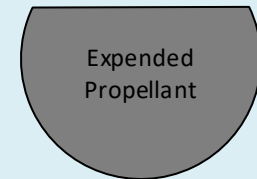


For example, GN2 (gaseous nitrogen) at 5 bar

At EOM, assume the residuals/hold-up are in the manifold, such that all of the propellant is expended from the tank.

C) Volume Change

- The difference between BOM and EOM volumes corresponds to the volume of the expended propellant, which can be found using the rocket and density equations
- For cold gas systems, this is calculated using the rocket equation
 - $V_{EOM} - V_{BOM} = V_{propellant}$



Approach

- Given that volume and pressure are inversely proportional for a constant gas (mass & temperature), the following equation can be derived:
 - $V_{EOM} / V_{propellant} = P_{BOM} / P_{BOM-EOM}$
- This can equation can be used to solve for the tank volume (V_{EOM}), which can then be used to solve for the pressurant mass & finally initial pressurant volume

Margin

- Tank capacity margin is applied to the propellant volume (rather than the final tank volume). This ensures that the calculations for pressurant include this additional tank volume.

Note that 296.8 is the specific gas constant for GN2 (J/kg K) and 300K is a standard temperature assumption.

Step 4: Determine Pressurant...

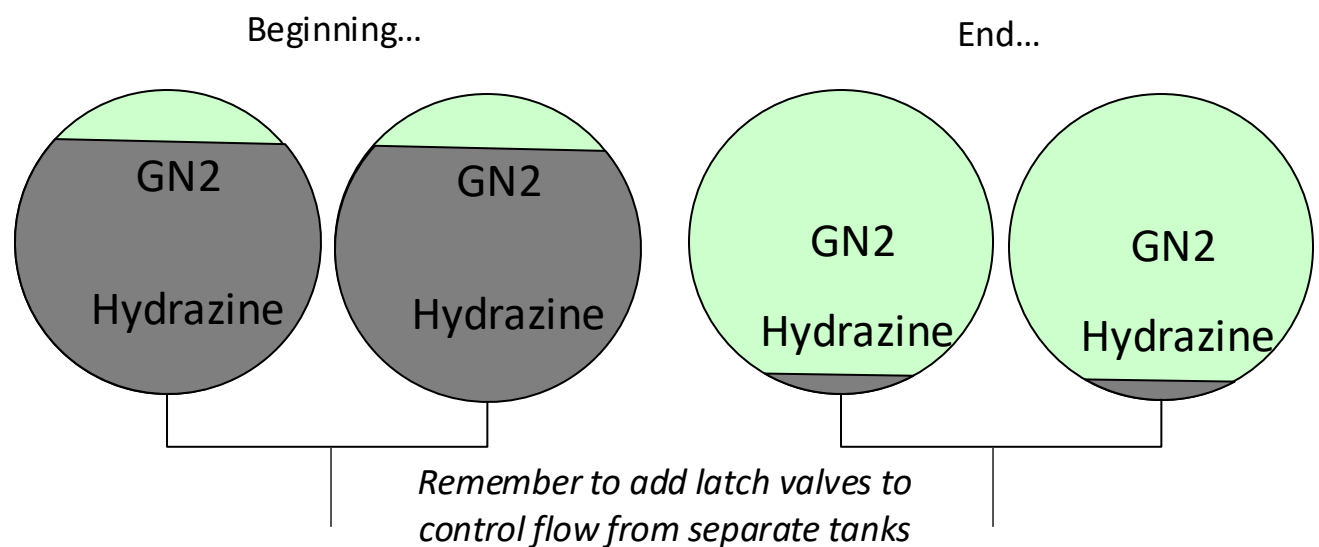
Propulsion

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- **Sizing**
- Assignment
- Hardware
- Integrated System
- Additional Info

• Other Combinations...

- Note that propellant is solved using the rocket equation, so the combinations below are slightly more complex (and require a balance between tanks)

Multiple Blowdown Tanks

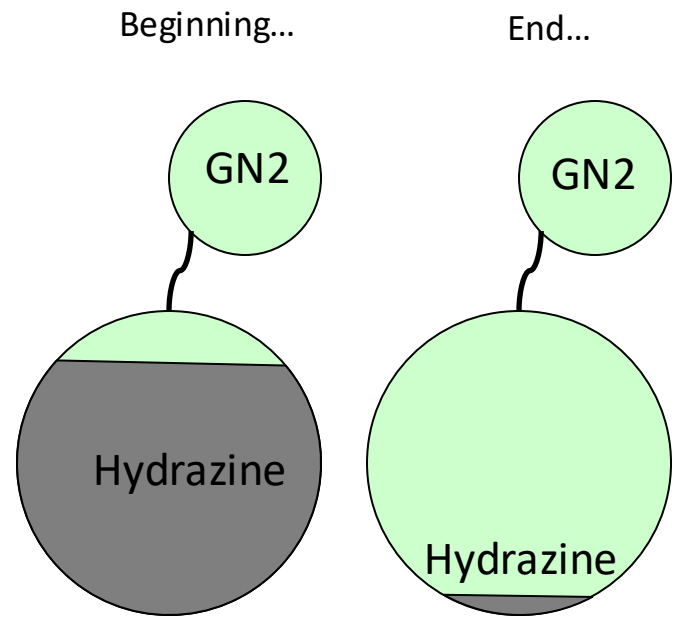


• When the ideal gas law breaks down (eg, Helium)

- Create an estimate using the ideal gas law ← sufficient for this class
- Use real gas relationships (Van der Waals) or multiply time an appropriate factor (eg, 1.2)

Regulated Tanks

(ie, separate pressurization system)



Step 5: Estimate Wet/Dry System Mass

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• **Sizing**

• Assignment

Hardware

Integrated System

Additional Info

• **Compile a mass list of all hardware in the schematic**

- Tanks*
- Thrusters*
- Pressure transducer, 0.3 kg
- Temperature sensor, 0.1 kg
- Latch valve, 0.7 kg
- Service valve, 0.25 kg
- System filter, 0.9 kg
- Cavitating venturi, 0.1 kg
- Accumulator, 0.25 kg (stores limited fuel quantities and/or stabilizes flow)
- Burst disk, 0.2 kg (provides pressure release [via rupture] when a specified threshold is reached)
- System tubing & fittings, 5% of above
- Mounting brackets, fasteners, etc., 5% of above

** Provided in homework and/or online search*

*Note that if you can't find a mass, use a guess/estimate and add a comment such as "**placeholder**"*

• **The values listed above are current best estimates (CBEs).**

- To these values, add contingency (ie, growth allowance) to accommodate for likely increases (tanks/thrusters 10%, valves/regulators/etc. 20%, tubing/brackets/etc. 30%)

• **System tubing/line estimates can be improved using a rough estimate of the spacecraft size and where tanks/thrusters would be located**

Step 5: Estimate Wet/Dry System Mass

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• **Sizing**

• Assignment

Hardware

Integrated System

Additional Info

- **One more: Solid Rocket Motor**
 - See [https://en.wikipedia.org/wiki/Star_\(rocket_stage\)](https://en.wikipedia.org/wiki/Star_(rocket_stage))
 - No additional propulsion system mass required
 - However, the use of an SRM needs to be coordinated with Structures, which needs to ensure an adequate supporting structure and (if required) a separation system
- **Solid Rocket Motors are used for one-time ΔV burns, where it can be more efficient and/or cost effective than a biprop system**
 - See the site above for total impulse capabilities for various star motors
 - Total Impulse capability = SRM Force (kN) x time (s)
 - Required impulse
 - Total Impulse required = Spacecraft wet mass x ΔV required
 - Typically, trajectories are designed for these to be equal. However, if there is a difference, this needs to be accommodated by a separate propulsion system



Step 5: Estimate Wet / Dry Mass

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• Sizing

• Assignment

Hardware

Integrated System

Additional Info

• Based on the design, create the corresponding dry mass & propellant tables

- As part of an exam question, you would just need to enter the component name (via pull-down) and quantity

Dry Mass

Component Name	Unit Mass kg	Quantity #	Mass kg	Contingency %	Total Mass kg	Comments
490-N Thruster	3.76	1	3.76	5%	3.95	
4-N Thrusters	0.37	10	3.70	5%	3.89	
Pressurant Tank	6.00	1	6.00	5%	6.30	NG-80558-1 (16 liters)
Fuel Tank	11.70	1	11.70	5%	12.29	NG-80394-1 (229 liters)
Oxidizer Tank	11.70	1	11.70	5%	12.29	NG-80394-1 (229 liters)
Pressure Transducers	0.30	12	3.60	5%	3.78	
Temperature Sensors	0.10	24	2.40	5%	2.52	
Pressure Regulator	1.10	2	2.20	5%	2.31	
Latch Valves	0.70	10	7.00	5%	7.35	
Service Valves	0.25	10	2.50	5%	2.63	
System Filters	0.90	10	9.00	5%	9.45	
Cavitating Venturi	0.10	2	0.20	5%	0.21	
System Tubes, Fittings			3.19	30%	4.14	5% of 2 above
Mounting brackets, Fasteners			3.35	30%	4.35	5% of 2 above
TOTAL					75.44	

Propellant Mass

Propellant	Propellant Mass kg	Comments
Pressurant	2.2	
Hydrazine	176.4	
Oxidizer	139.3	

ACS Thruster Design & Placement

Propulsion

Introduction

Background

Application

- Requirements
- Architecture
- Schematic
- **Sizing**
- Assignment

Hardware

Integrated System

Additional Info

- **Often called a Reaction Control System (RCS) when used only for attitude control**
 - From propulsion, these are typically the smallest thrusters. Often, they functionally back-up a reaction wheels in case of a failure, but they do not provide the same precision
- **Characteristics**
 - Performance: 0.1-5.0 deg pointing control
 - Various sizes available, with different levels of performance (ie, control)
 - Propellant-limited and may introduce contamination
 - Capable of very high torque
 - Accuracy limited by thruster configuration (impulse bit)



ACS Thruster Design & Placement

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• Sizing

• Assignment

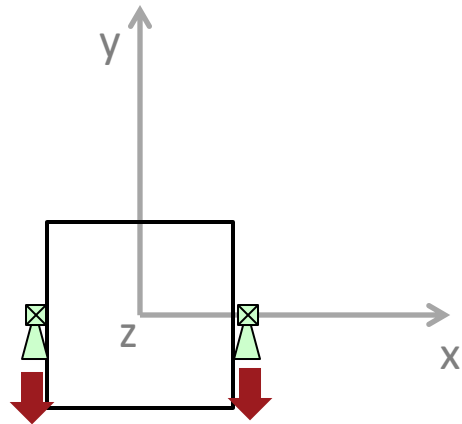
Hardware

Integrated System

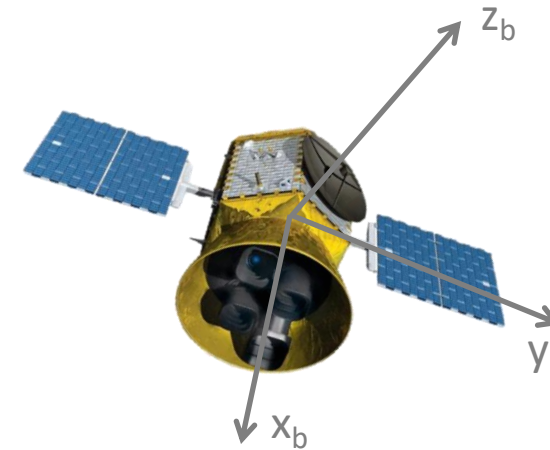
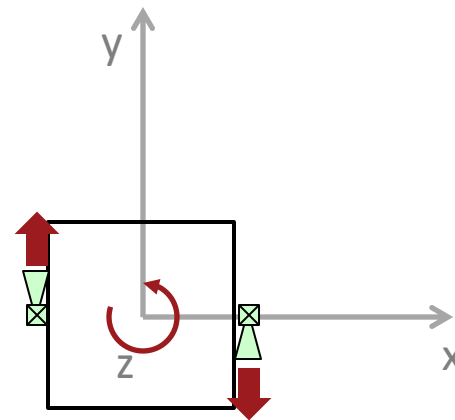
Additional Info

• RCS Thruster control?

1-DOF translational (+y-axis)



1-DOF rotational (+z-axis)



• The above graphic shows “coupled” thrusters that act through the center of mass (CM)

- Thus, 1-DOF = two coupled thrusters
- This is not required, but results in much simpler operations & improved control

• Redundancy

- Dual-redundancy typically means two strings (A & B)
- Thus, 1-DOF = four coupled, redundant thrusters (A-string, B-string)

• Thrusters are typically placed

- On opposite sides of the CM as shown above
- On the outer edge (or beyond it) to provide greater control for a longer moment-arm
- Typically in clusters of 2-4 thrusters that face different directions

RCS Thruster Configuration

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• Sizing

• Assignment

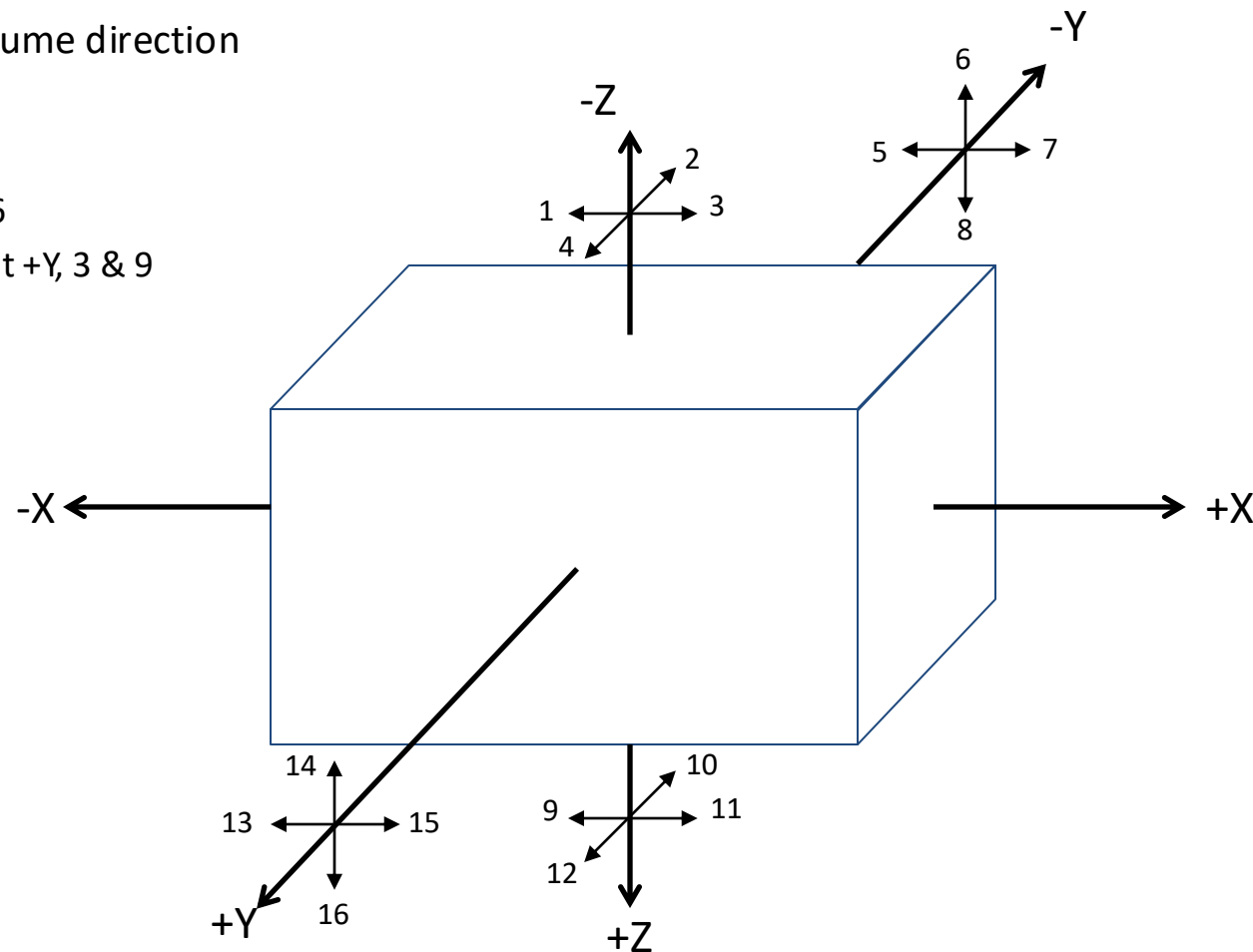
Hardware

Integrated System

Additional Info

• Notional diagram for a set of 16 thrusters in 4 clusters

- 6 DOF thruster configuration
- Arrows indicate plume direction
- Examples
 - For +X, fire 1 & 9
 - For -Z, fire 8 & 16
 - For rotation about +Y, 3 & 9



RCS Thruster Configuration

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• Sizing

• Assignment

Hardware

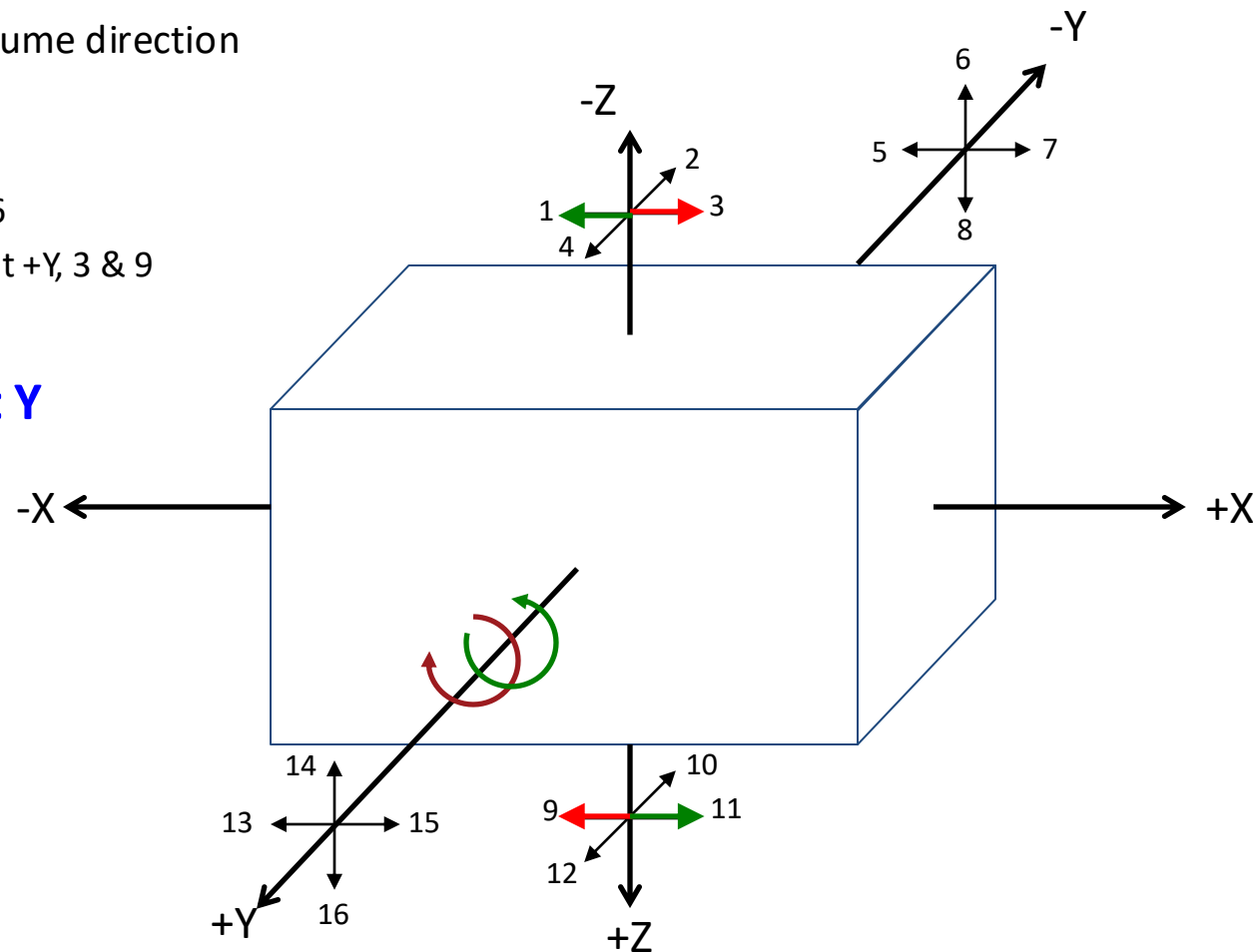
Integrated System

Additional Info

• Notional diagram for a set of 16 thrusters in 4 clusters

- 6 DOF thruster configuration
- Arrows indicate plume direction
- Examples
 - For +X, fire 1 & 9
 - For -Z, fire 8 & 16
 - For rotation about +Y, 3 & 9

Rotations about Y



RCS Thruster Configuration

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• Sizing

• Assignment

Hardware

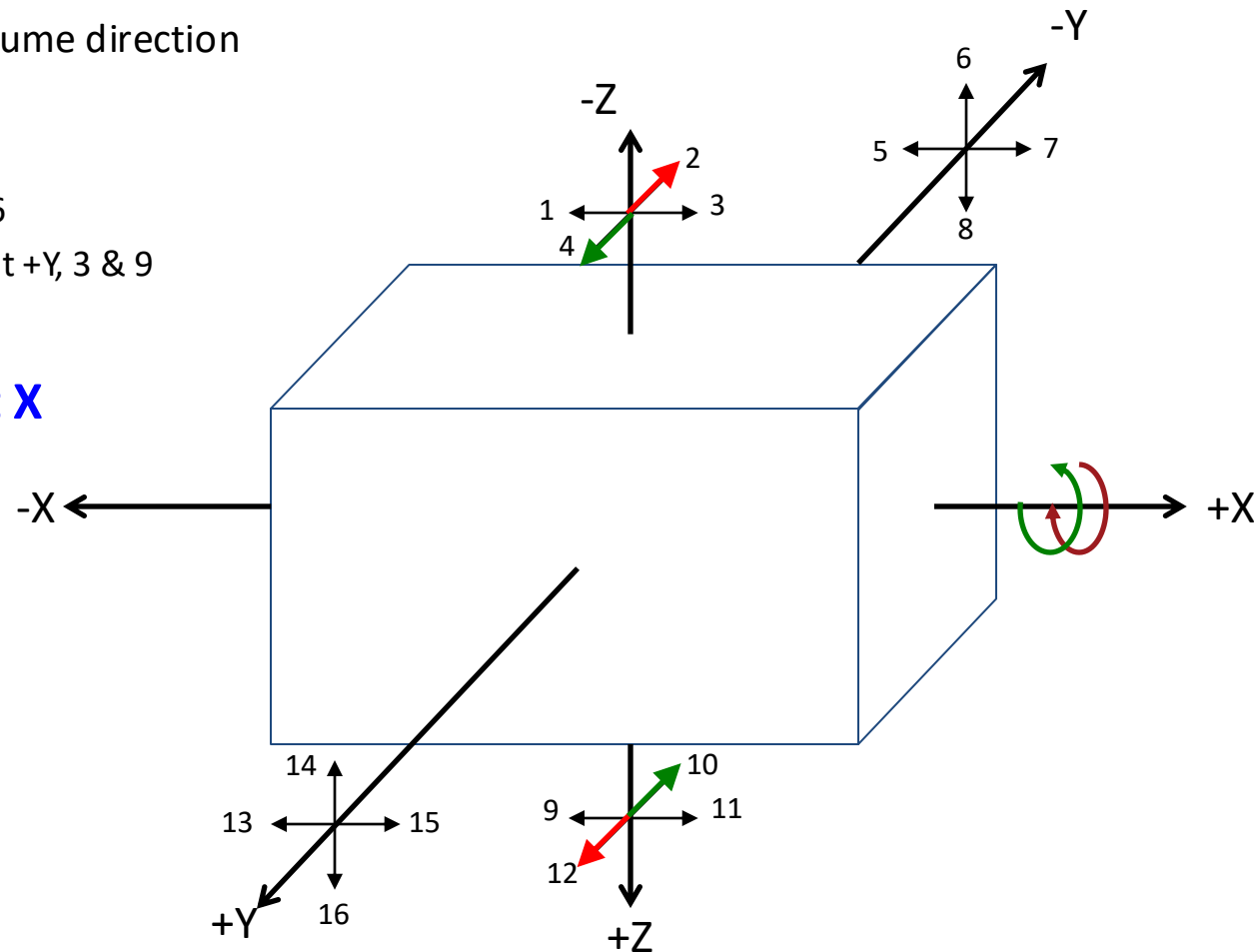
Integrated System

Additional Info

• Notional diagram for a set of 16 thrusters in 4 clusters

- 6 DOF thruster configuration
- Arrows indicate plume direction
- Examples
 - For +X, fire 1 & 9
 - For -Z, fire 8 & 16
 - For rotation about +Y, 3 & 9

Rotations about X



RCS Thruster Configuration

Propulsion

Introduction

Background

Application

• Requirements

• Architecture

• Schematic

• Sizing

• Assignment

Hardware

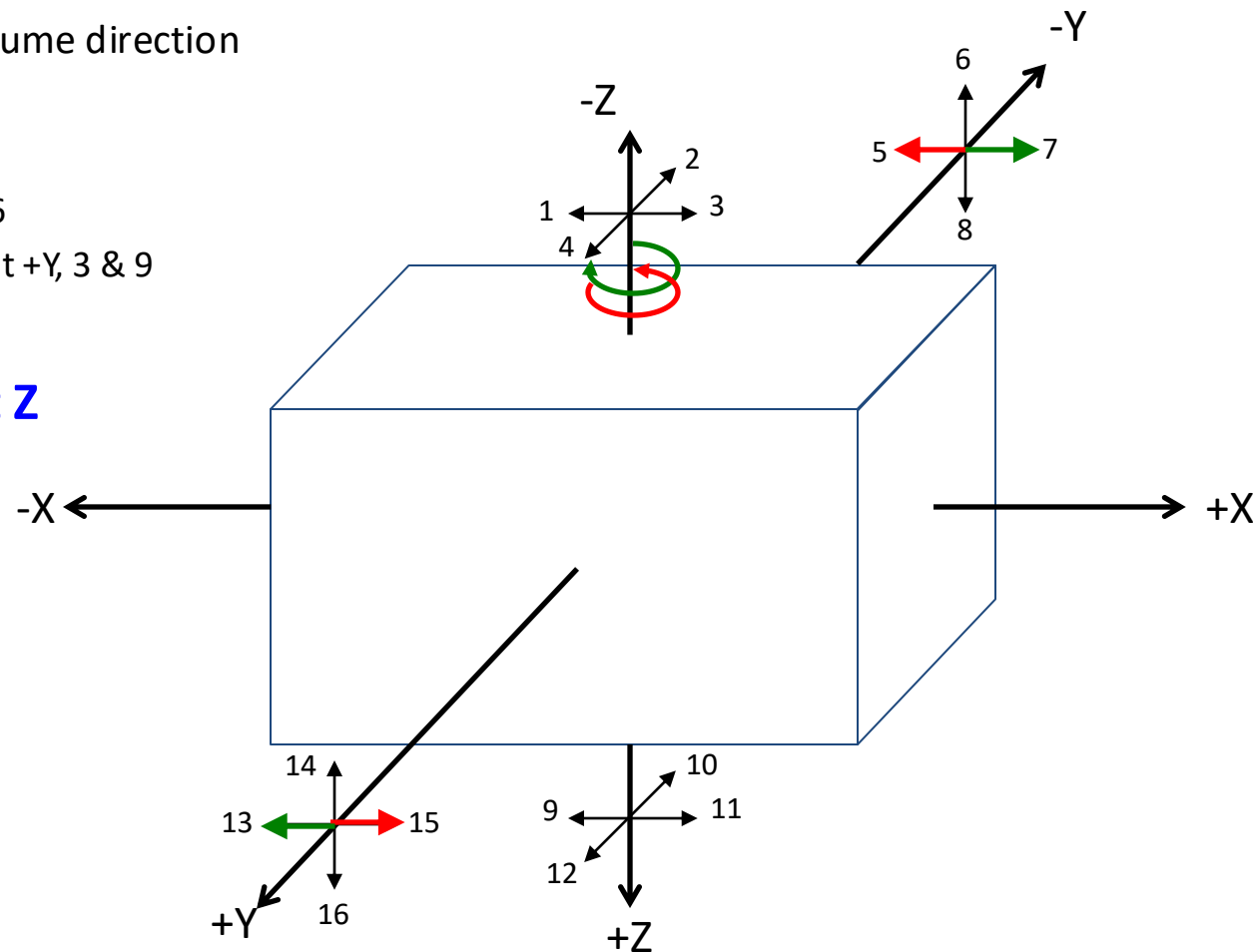
Integrated System

Additional Info

• Notional diagram for a set of 16 thrusters in 4 clusters

- 6 DOF thruster configuration
- Arrows indicate plume direction
- Examples
 - For +X, fire 1 & 9
 - For -Z, fire 8 & 16
 - For rotation about +Y, 3 & 9

Rotations about Z



Thrusters (1 of 3)

Propulsion

Introduction

Background

Application

Hardware

• **Components**

• MEL

• Rqmts & Margin

• Assignment

Integrated System

Additional Info

- **Primary Thruster Characteristics**

- Thrust-level (eg, 1-N vs. 490-N), which determines the ability to provide thrust and thus ΔV
- Efficiency (ie, I_{sp}), which is based on both type of thruster & propellant
- Minimum impulse bit (MIB) is the smallest control torque that can be applied to the spacecraft

- **Primary Uses**

- Trajectory Correct Maneuvers (TCMs) for changing ΔV
- Orbital Maintenance
- Momentum dumps
- Attitude (or reaction) control
- Thrust vector control (TVC) that provide guidance on larger maneuvers

- **Matching Thrusters to Spacecraft**

- Consider necessary functions: large ΔV maneuvers, attitude control, etc.
- Thrusters should be evaluated for min/max burn times
 - Minimum burn time should be consistent with the minimum impulse bit
 - Maximum burn time should be consistent with the type of maneuver (eg, orbit insertion, burn time < 2-3 hrs)
- Result is often something like...
 - RCS thrusters for attitude control, momentum dumps, etc.
 - Main engine(s) for large & more efficient ΔV maneuvers
 - Other mid-size thrusters where the above thrusters aren't sufficient (eg, thrust vector control on a large ΔV burn)
- Reminder that thrust-level can be scaled with the number of thrusters

Thrusters (2 of 3)

Propulsion

Introduction

Background

Application

Hardware

• Components

• MEL

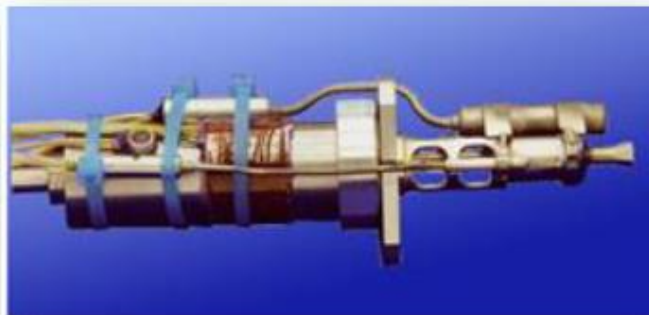
• Rqmts & Margin

• Assignment

Integrated System

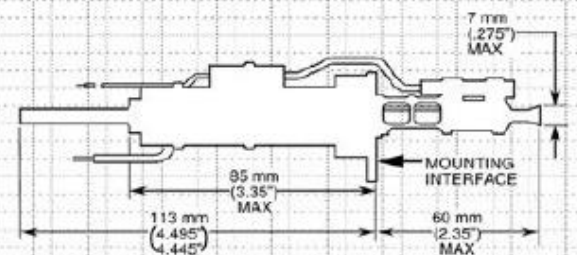
Additional Info

MR-103G 1N (0.2 lbf) Rocket Engine Assembly



P/N 34308-303

ICD 34309



Design Characteristics

• Propellant.....	Hydrazine
• Catalyst.....	S-405
• Thrust/Steady State.....	1.13 - 0.19 N (0.253 - 0.043 lbf)
• Feed Pressure.....	28.3 - 4.8 bar (420 - 70 psia)
• Chamber Pressure.....	23.8 - 4.5 bar (345 - 65 psia)
• Expansion Ratio.....	100:1
• Flow Rate.....	0.5 - 0.09 g/sec (0.0011 - 0.0002 lbm/sec)
• Valve.....	Dual Seat
• Valve Power.....	8.25 Watts Max @ 28 Vdc & 21°C
• Valve Heater Power.....	1.54 Watts Max @ 28 Vdc & 21°C
• Cat. Bed Heater Pwr.....	6.32 Watts Max @ 28 Vdc & 21°C
• Mass.....	0.33 kg (0.73 lbm)
Engine.....	0.13 kg (0.24 lbm)
Valve.....	0.20 kg (0.44 lbm)
Heaters.....	0.033 kg (0.14 lbm)

Performance

• Specific Impulse.....	224 - 202 sec (lbf-sec/lbm)
• Total Impulse.....	97,078 N-sec (21,825 lbf-sec)
• Total Pulses.....	835,017
• Min Impulse Bit.....	0.0133 N-sec @ 6.9 bar & 15 ms ON (0.003 lbf-sec @ 100 psia & 15 ms ON)
• Steady State Firing.....	300 & 1,000 sec Single Firing 23.8 — 40.6 hrs Cumulative

Status

- Flight Proven
- Currently in Production; Transitioning to MR-103J

Reference

- AIAA-2005-3952

Thrusters (3 of 3)

Propulsion

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Hardware

• **Components**

• MEL

• Rqmts & Margin

• Assignment

Integrated System

Additional Info

Cold Gas Thrusters	Thrust N	Mass kg	Nom. Pres. bar	MEOP bar	Heritage
0.01-N Thruster	0.01	0.04	1.5	10	CHAMP, GRACE
0.04-N Thruster	0.04	0.04	1.5	10	CHAMP, GRACE
0.12-N Thruster	0.12	0.14	6.9	20.7	SIRTF
3.6-N Thruster	3.6	0.23	15.7	15.7	SAFER, SCIT, Pluto Fast
1-N Thruster	1.0	0.115	90	186	GEO applications

Loosely based on...

<https://www.moog.com/products/propulsion-controls/spacecraft/thrusters.html>

Monoprop Thrusters	N	kg	bar	bar	Comments
1-N MR-103G thruster (monoprop)	1	0.33	4.8	28.3	Aerojet, flight proven
4-N MR-111G thruster (monoprop)	4	0.37	6.7	24.1	Aerojet, flight proven
22-N MR-106L thruster (monoprop)	22	0.59	5.9	27.6	Aerojet, flight proven
110-N MR-107T thruster (monoprop)	110	1.01	7.0	35.0	Aerojet, flight proven
300-N MR-107U thruster (monoprop)	300	1.38	20.6	52.4	Aerojet, flight proven
3,100-N MR-80B thruster (monoprop)	3100	168.0	47.2	47.2	Aerojet, flight proven

Source

<https://www.rocket.com/space/space-power-propulsion/monopropellant-rocket-engines>

Monoprop Thrusters	Thrust N	Mass kg	Min Pressure bar	Op. Pressure bar	Comments
22-N R-6F thruster (biprop)	22	0.97	6.9	20.79	MMH/NTO, Aerojet, flight proven
110-N R-1E thruster (biprop)	110	2.00	6.9	27.6	MMH/NTO, Aerojet, flight proven
490-N R-4D-11 thruster (biprop)	490	3.76	4.1	29.3	MMH/NTO, Aerojet, flight proven
890-N R-42DM thruster (dual-mode)	890	7.3	5.5	31.0	Hydrazine, Aerojet, flight proven
4000-N R-40B thruster (biprop)	4000	10.5	10.3	27.6	MMH/NTO, Aerojet, flight proven

Source

<https://www.rocket.com/space/space-power-propulsion/monopropellant-rocket-engines>

Thrusters (3 of 3)

Propulsion

Introduction

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Hardware

• **Components**

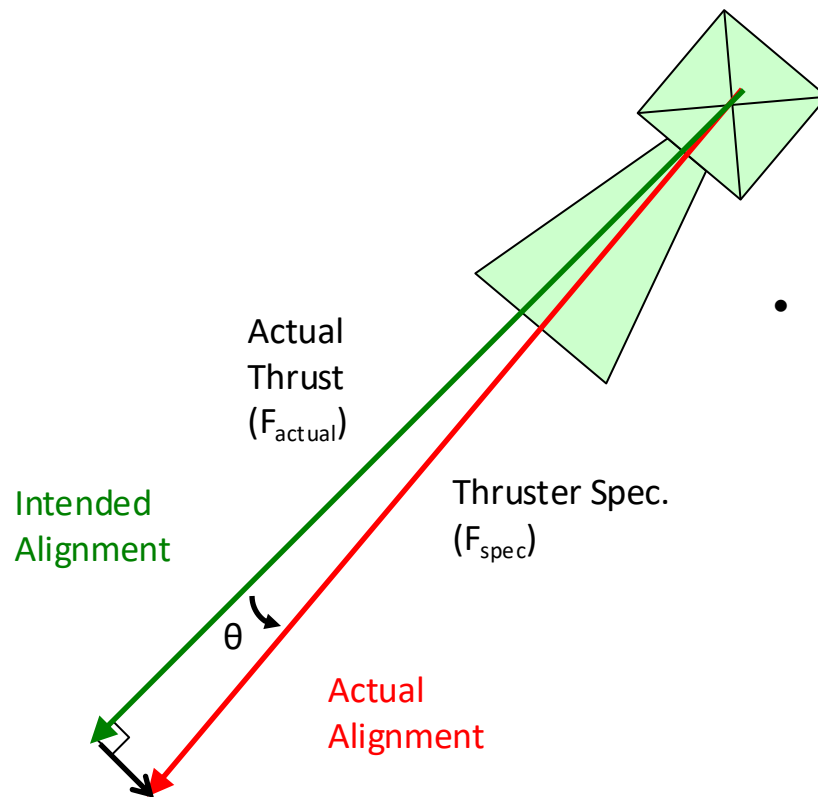
• MEL

• Rqmts & Margin

• Assignment

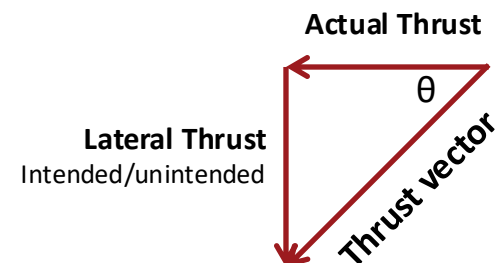
Integrated System

Additional Info



Misalignment Angle (when unintended)

Cant Angle (when intended)



Check Your Work

0-deg: Thrust = actual thrust

90-deg: Zero thrust (burn-time is infinite)

- **When the a thrust vector is misaligned or canted, there is a “cosine loss” that needs to be accounted for**

– The equation is:

$$F_{\text{actual}} = F_{\text{spec}} \times \text{Cosine}(\theta)$$

– Example for a 5N and 10-deg cant angle

$$5 \text{ N} \times \text{COS}(10\text{-deg}) = 4.92 \text{ N}$$

When solving for what size thruster is needed, the cosine term is divided from the required value (ie, slightly more thrust is needed due to this effect).

Note that the reason to cant thrusters (when coupled) is that it provides additional lateral stability. If not coupled or if the alignments are not equal, this introduces an error.

Thruster Stability & Duration

Propulsion

Introduction

Background

Application

Hardware

• **Components**

• MEL

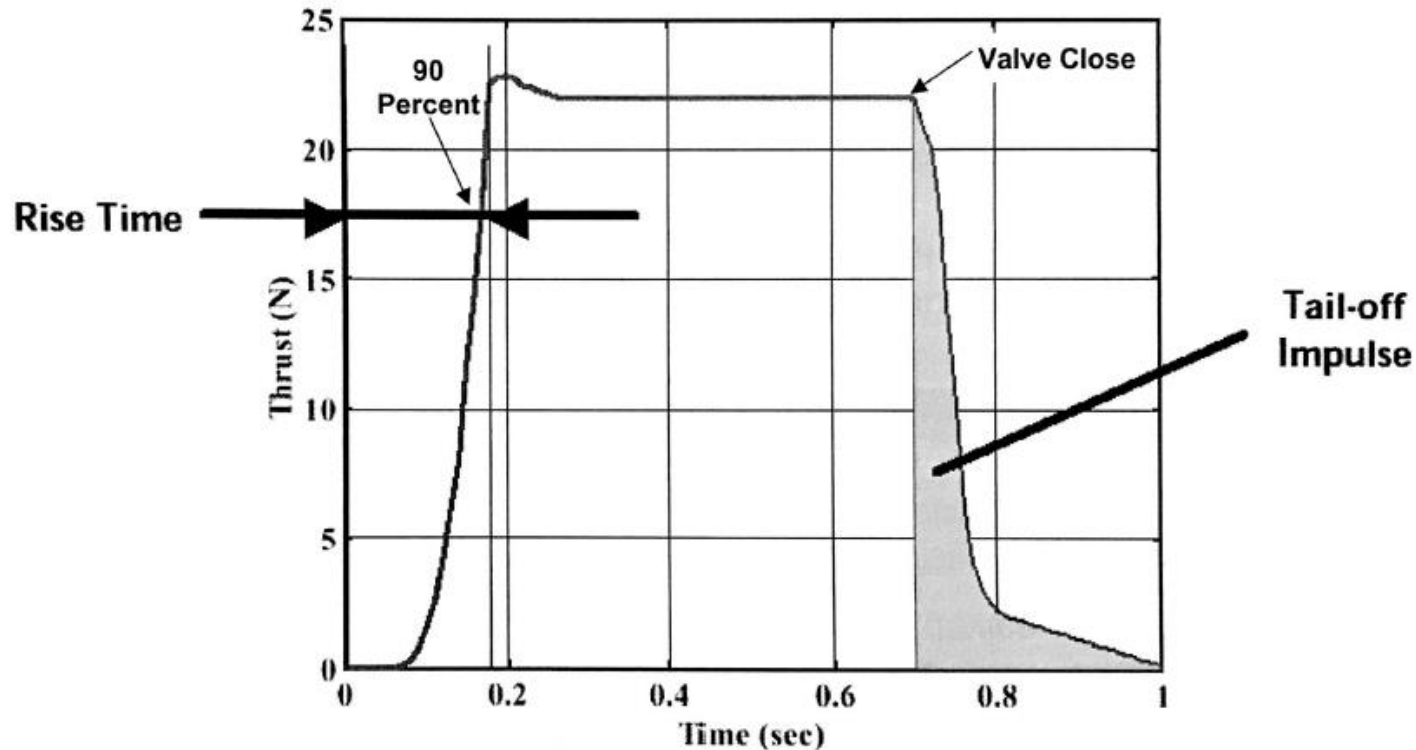
• Rqmts & Margin

• Assignment

Integrated System

Additional Info

- The graph below shows a typical thruster firing
 - The Rise Time & Tail-Off result in slight inefficiencies
 - Additionally, the minimum duration thruster firing is known as the minimum impulse bit. This determines the resolution/control of thruster firings, which can drive both maneuver and attitude control
 - Note that a thruster may be fired in steady state or pulse mode.



Propulsion

Introduction

Background

Application

Hardware

• **Components**

• MEL

• Rqmts & Margin

• Assignment

Integrated System

Additional Info

Propellant Tanks

IHIAeroSpace

IHI Aerospace (IA) is one of the major manufacturers of tanks and propulsion systems in Japan and has manufactured a large number of tanks since its founding in 1964. IA's product line covers a wide range of tanks from 3 to 1494 liters in volume and from diaphragms to surface tension channel mesh propellant management devices (PMDs).

Design

- Stress analysis (including PMDs)
- Dynamic analysis
- Leak before burst (LBB) and flow growth analysis
- Analysis of the propellant behavior

Standards

- MIL-STD-1522A
- JAXA standards, NASA standards, ESA standards



430 L tank for SELENE



Channel Mesh



Vane



Inner Vane



Vent Hole at Circumferential position

Drop tower test



552 L tanks for HTV



Diaphragm

Tanks (2 of 3)

Propulsion

- Introduction
- Background
- Application
- Hardware
- **Components**
- MEL
- Rqmts & Margin
- Assignment
- Integrated System
- Additional Info

Cold Gas Tank List	Volume	Tank Mass	MEOP (Op Pressure)	
Name	liters	kg	bar	Program
NG-80615-1	28	6.8	310.3	
NG-80412-1	50	7	150	ETS8; Xenon
NG-80400-1	67.3	10	310	A2100
NG-80446-1	67.3	10.7	310	A2100; 80446; Qualified E
NG-80525-1	67.3	10.7	310	IRNS; Pressurant Tank. Th
NG-80436-1	81.4	12.7	331	ASTROLINK; 80436; Quali
NG-80496-1	81.4	12.7	331	DS2000; Pressurant Tank.
NG-80475-201	87	16.8	310	
NG-80592-1	129.8	20.5	185.1	
NG-80326-1	3.9	1.5	248	MAGELLAN; 80326; Quali
NG-80195-1	9.4	5.4	184	MARINER
NG-80558-1	16	5.99	175	
NG-80314-201	36.1	16	248	TRMM; 80314; Qualified I
NG-80510-1	36.6	11.9	335.8	

Source
 See 331a_PropTankData_20200908, along with
<https://www.northropgrumman.com/space-old/pressurant-tanks-data-sheets-sorted-by-volume/>

Tanks (3 of 3)

Propulsion

- Introduction
- Background
- Application
- Hardware
- **Components**
- MEL
- Rqmts & Margin
- Assignment
- Integrated System
- Additional Info

Liquid Prop Tank List	Volume	Tank Mass	MEOP (Op Pressure)	
Name	liters	kg	bar	Comments
NG-80568-1	3.8	1.36	24	LMP-103; HPGP
NG-80296-1	58.6	3.6	29.3	
NG-80599	94.6	6.6	22	; 80387; Qualified En
NG-80462-1	161.5	10	29	80462; Protoflight Er
NG-80394-1	229.2	11.7	20.7	
NG-80561-1	311	15.88	20.7	Ox
NG-80373-1	337.4	13.8	20.7	; 80373; Qualified En
NG-80317-105	383.6	27.2	24.1	
NG-80340-1	503.8	17.1	19	
NG-80484-1	646.6	23.4	20.7	80484; Protoflight Er
NG-80432-1	858	30.7	20.7	Fuel; Derivative of P/
NG-80425-101	938	33.1	20.7	80425; Qualified Env
NG-80501-1	1025.4	35.4	16	
NG-80404-5	1196	37.51	10.34	
NG-80441-1	1275.5	41.3	17.2	
NG-80528-1	1511.68	55.34	20.7	Fuel
NG-80554-1	1754	61.2	17.2	

Source

See 331a_PropTankData_20200908, along with
<https://www.northropgrumman.com/space-old/pressurant-tanks-data-sheets-sorted-by-volume/>

Solid Rocket Motors (SRMs)

Propulsion

Introduction

Background

Application

Hardware

• **Components**

• MEL

• Rqmts & Margin

• Assignment

Integrated System

Additional Info

STAR Motor	Dry Mass kg	Prop Mass kg	Specific Impulse s	Total Impulse kNs
Star-3A (0.29 kNs)	0.7	0.12	241	0.3
Star-3 (1.25 kNs)	0.7	0.48	266	1.3
Star-5A (5.73 kNs)	2.4	2.27	250.8	5.7
Star-12 (46 kNs)	9.4	18.28	252	46.0
Star-13 (86.6 kNs)	4.6	30.98	273	83.6
Star-17 (198 kNs)	9.5	69.63	286.2	197.9
Star-24 (560 kNs)	18.33	199.90	282.9	560.5
Star-24C (613 kNs)	19.73	219.50	282.3	613.9
Star-37 (1584 kNs)	62.7	558.40	260	1584.46
Star-37N (1590 kNs)	63.5	559.30	290	1590.24
Star-37B (1858 kNs)	64.7	653.70	291	1858.91
Star-37D (1858 kNs)	64.7	653.70	266	1858.91
Star-37S (1872 kNs)	53.4	658.00	287.3	1872.43
Star-37XF (2342 kNs)	67.1	815.40	291.1	2342.74
Star-37F (2444 kNs)	67.3	866.80	286	2444.46
Star-37XFP (2537 kNs)	71.7	883.60	290	2537.49
Star-37XF (2542 kNs)	67.7	885.40	290	2542.03
Star-37C (2707 kNs)	82.8	964.70	285.5	2707.19
Star-37E (2910 kNs)	83.1	1039.60	283.6	2910.03
Star-37G (2988 kNs)	86.4	1065.90	289.9	2988.36
Star-37X (3047 kNs)	82.8	1067.20	295.6	3047.69
Star-37FM (3051 kNs)	81.5	1065.90	289.8	3051.35
Star-37Y (3118 kNs)	80.6	1071.40	297	3118.2

Source

[https://en.wikipedia.org/wiki/Star_\(rocket_stage\)](https://en.wikipedia.org/wiki/Star_(rocket_stage))

<https://www.northropgrumman.com/wp-content/uploads/NG-Propulsion-Products-Catalog.pdf>

STAR 48A TE-M-799



LONG NOZZLE

The STAR 48A motor is designed as an increased payload capability version of the basic STAR 48 by incorporating an 8-inch stretch of the motor case. The long nozzle version maximizes performance by also incorporating an 8-inch-longer exit cone, resulting in a longer overall envelope.

The design uses a high-energy propellant and high-strength titanium case. The submerged nozzle uses a carbon-phenolic exit cone and a 3-D carbon-carbon throat.

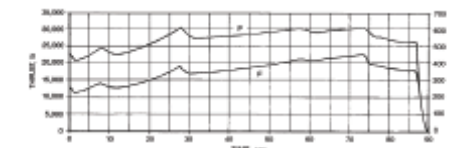
The case features forward and aft mounting flanges and multiple tabs for attaching external hardware that can be relocated or modified for varying applications without requalification.

Disconti

Qualifie

Disconti

Develop



NORTHROP
GRUMMAN

MOTOR DIMENSIONS	
Motor diameter, in.	48.0
Motor length, in.	88.0
MOTOR PERFORMANCE (75°F VACUUM)	
Burn time/section time, sec.	87.2/88.2
Ignition delay time, sec.	0.100
Burn time average chamber pressure, psia	543
Maximum chamber pressure, psia	607
Total impulse, lbf-sec.	1,563,790
Propellant specific impulse, lbf-sec/lbm.	291.9
Effective specific impulse, lbf-sec/lbm.	289.9
Burn time average thrust, lbf	17,750
Maximum thrust, lbf	21,050
NOZZLE	
Initial throat diameter, in.	4.49
Exit diameter, in.	29.5
Expansion ratio, initial	43.1:1
WEIGHTS, lbm	
Total loaded*	5,891.1
Propellant (including igniter propellant)	5,357.2
Case assembly	153.8
Nozzle assembly (excluding igniter propellant)	101.8
Total inert	333.9
Burnout*	294.3
Propellant mass fraction*	0.94
*Excluding remote B&A/ETA	
TEMPERATURE LIMITS	
Operation	30°-100°F
Storage	30°-100°F
SPIN EXPERIENCE, RPM	80
PROPELLANT DESIGNATION	TP-H-3340
CASE MATERIAL	TITANIUM
PRODUCTION STATUS	DEVELOPMENT

For more information, contact:
starmotors@ngc.com
northropgrumman.com

Propellant Management Devices

Propulsion

- Introduction
- Background
- Application
- Hardware
- Components**
- MEL
- Rqmts & Margin
- Assignment
- Integrated System
- Additional Info

- While diaphragms are often used to prevent mixing between the propellant and pressurant, there also may be no barrier as well, and the propellant liquid is controlled via vanes and other devices (based on liquid surface tension)

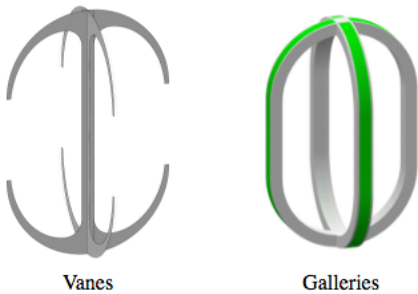
PMD Types

Vanes	Galleries	Sponges	Traps	Troughs	Porous Elements	Combinations
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Propellant Management Devices, or PMDs, are static all metal structures inside spacecraft propellant tanks that use surface tension to ensure gas free liquid delivery to the tank outlet. They can be made entirely from titanium allowing their use in the most corrosive propellants. With no moving or deforming parts, they are inherently reliable.

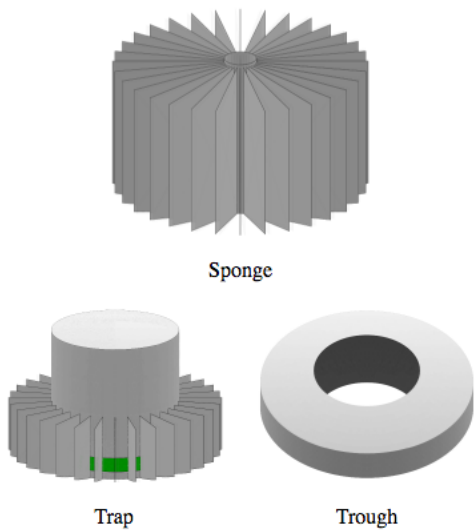
Communication Devices

Communication PMDs provide a flow path from the propellant pool to the outlet. They can provide unlimited demand in any direction and, therefore, are the most flexible of PMDs. Unfortunately, if the acceleration level precludes vanes, galleries are required. Galleries are the heaviest, the most costly, and, importantly, the least reliable of PMD components (but also the most capable).



Control Devices

Control PMDs retain some or all of the tank's propellant over the outlet. They provide for limited duration demand or control the propellant CG. They may retain propellant for repetitious maneuvers - refilling between maneuvers. Alternatively, they can provide propellant for once in a lifetime maneuvers such as spin recovery or a station change maneuver. Sponges, Traps and Troughs are control devices.



Propulsion

Introduction

Background

Application

Hardware

• Components

• MEL

• Rqmts & Margin

• Assignment

Integrated System

Additional Info

- Schematic
- Master Equipment List (MEL)
- Propellant budget
- Power analysis (next semester)
- Cost estimate (next semester)

Loosely related examples...

Mass Summary	CBE	Contingency
	kg	%
Flight System	124.9	17
Payload	10.0	50
Spacecraft Bus	114.9	14
Propulsion	41.0	7
ACS	5.2	5
Telecom	5.2	10
C&DH	6.6	7
Power	30.0	14
Thermal	3.9	30
Structure	23.0	30
Launch Vehicle Summary	MEV	Ma
Spacecraft 1 Wet Mass	980.4	5
Dry Mass	146.2	20
Propellant	834.3	2
Spacecraft 2 Wet Mass	980.4	5
Dry Mass	146.2	20
Propellant	834.3	2
Total Wet Mass	1960.9	5
Launch Vehicle Reserve		
Launch Vehicle Allocation		

Mass Table	Unit (kg)	#	CBE
Architecture 1: HW-1 example		Total	41.0 kg
22-N Thruster	1.10	1	1.1
0.8 N Thrusters	0.10	8	0.8
Warm Gas Generator	3.00	1	3.0
Propellant Tank (NG-80425-101)	33.10	1	33.1
Misc. Hardware	3.00	1	3.0

Dry Mass

Component Name	Unit Mass kg	Quantity #	Mass kg	Contingency %	Total Mass kg	Comments
490-N Thruster	3.76	1	3.76	5%	3.95	
4-N Thrusters	0.37	10	3.70	5%	3.89	
Pressurant Tank	6.00	1	6.00	5%	6.30	NG-80558-1 (16 liters)
Fuel Tank	11.70	1	11.70	5%	12.29	NG-80394-1 (229 liters)
Oxidizer Tank	11.70	1	11.70	5%	12.29	NG-80394-1 (229 liters)
Pressure Transducers	0.30	12	3.60	5%	3.78	
Temperature Sensors	0.10	24	2.40	5%	2.52	
Pressure Regulator	1.10	2	2.20	5%	2.31	
Latch Valves	0.70	10	7.00	5%	7.35	
Service Valves	0.25	10	2.50	5%	2.63	
System Filters	0.90	10	9.00	5%	9.45	
Cavitating Venturi	0.10	2	0.20	5%	0.21	
System Tubes, Fittings			3.19	30%	4.14	5% of above
Mounting Brackets, Fasteners			3.35	30%	4.35	5% of above
TOTAL					75.44	

Propellant Mass

Propellant	Propellant Mass kg	Comments
Pressurant	2.2	
Hydrazine	176.4	
Oxidizer	139.3	

Example Requirements

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Additional Info

- **Flight System**
 - The total flight system mass shall be less 500 kg
- **Orbital Mechanics**
 - The flight system shall provide 500 m/s of ΔV capability post-launch
 - The flight system shall support a maximum ΔV burn of 400 m/s
 - Or, flight system shall support the trajectory ΔV budget in Table X-X
- **Attitude Control**
 - The propulsion system shall be capable of 3-axis control while orbiting Mars
 - The propulsion system shall provide a +X axis spin of 1.0 rpm
- **Propellant**
 - The propellant tanks shall store 100 to 110 kg of propellant
- **Thrusters**
 - The main engine shall be capable of a maximum duration burn of 40 min
- **Pressurization**
 - The pressurization system shall maintain a 400 psia for Mars Orbit Insertion (JOI)
- **Redundancy**
 - The propulsion system shall be fully redundant
- **Contamination Control**
 - The propulsion system shall not release hydrazine contaminants
- **Note**
 - “should” = you don’t really have to...
 - For a goal, you would say.. The spacecraft shall X, with an objective [or goal] of Y

Example Requirements

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Additional Info

• Flight System

- The total flight system mass shall be less 500 kg

• Orbital Mechanics

- The flight system shall provide 500 m/s of ΔV capability post-launch
- The flight system shall support a maximum ΔV burn of 400 m/s
 - Or, flight system shall support the trajectory ΔV budget in Table X-X

• Attitude Control

- The propulsion system shall be capable of 3-axis control while orbiting Mars
- The propulsion system shall provide a +X axis spin of 1.0 rpm

• Propellant

- The propellant tanks shall store 100 to 110 kg of propellant

• Thrusters

- The main engine shall be capable of a maximum duration burn of 40 min

• Pressurization

- The pressurization system shall maintain a 400 psia for Mars Orbit Insertion (MOI)

• Redundancy

- The propulsion system shall be fully redundant

• Contamination Control

- The propulsion system shall not release hydrazine contaminants

Be able to rewrite the rocket equation to use dry mass or wet mass

Use ΔV and mass to determine the required propellant & pressurant

More of an ACS question, but determine the number & placement of thrusters

Be able to incorporate lower-level requirements/constraints in a design

Size thrusters based on pressure & burn duration

Apply redundancy throughout the system

Consider alternative designs or propellants for specific constraints

Interpreting Schematics

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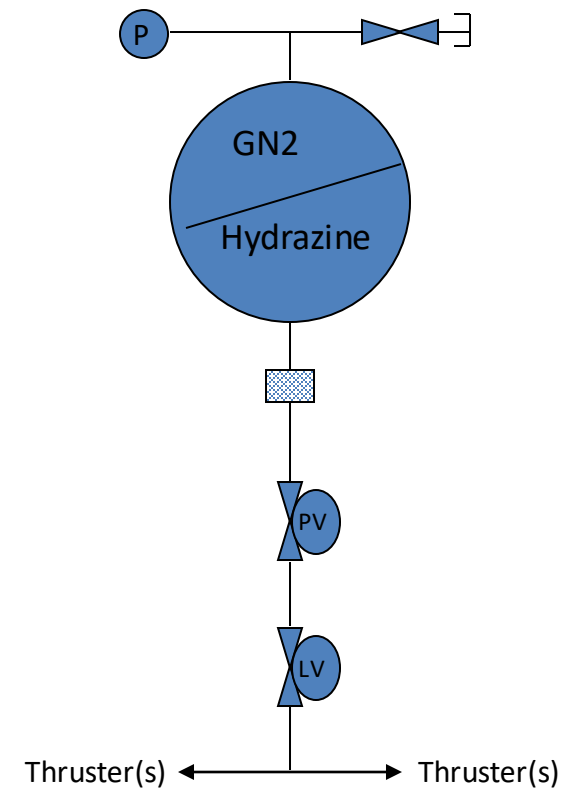
• Trade Studies

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Additional Info

- **What are the components?**
- **What type of propulsion system is this?**
- **What is the mission duration?**
 - Short (≤ 1 year) versus long (> 2 years)
- **What's wrong with this design?**



Single-string, RCS

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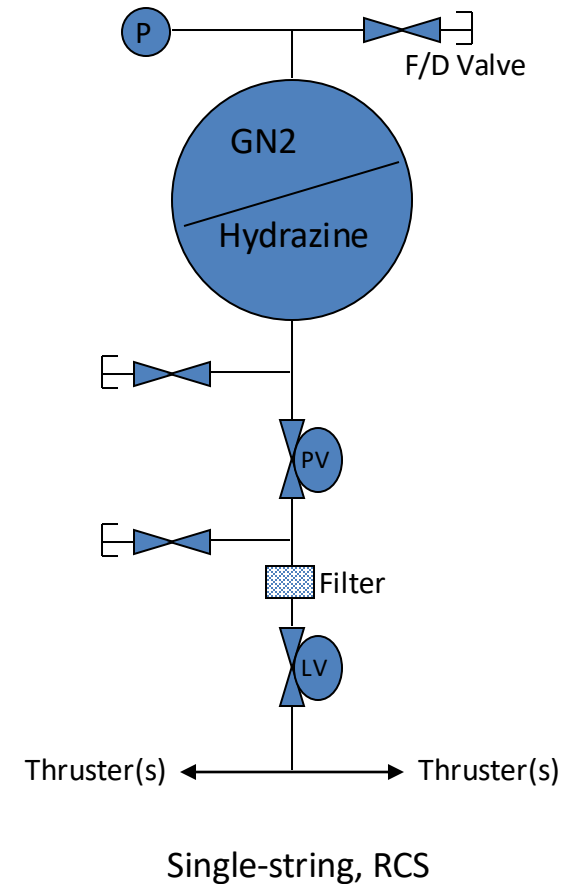
• Trade Studies

• Case Study

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Additional Info

- **What are the components?**
- **What type of propulsion system is this?**
 - Simple monoprop blowdown hydrazine system
- **What is the mission duration?**
 - **Short (≤ 1 year)** versus long (> 2 years)
- **What's wrong with this design?**
 - No filter downstream of the pyro valve
 - No service valve below monoprop tank
 - No service valve downstream of pyrovalve to test the latch valve



Interpreting Schematics

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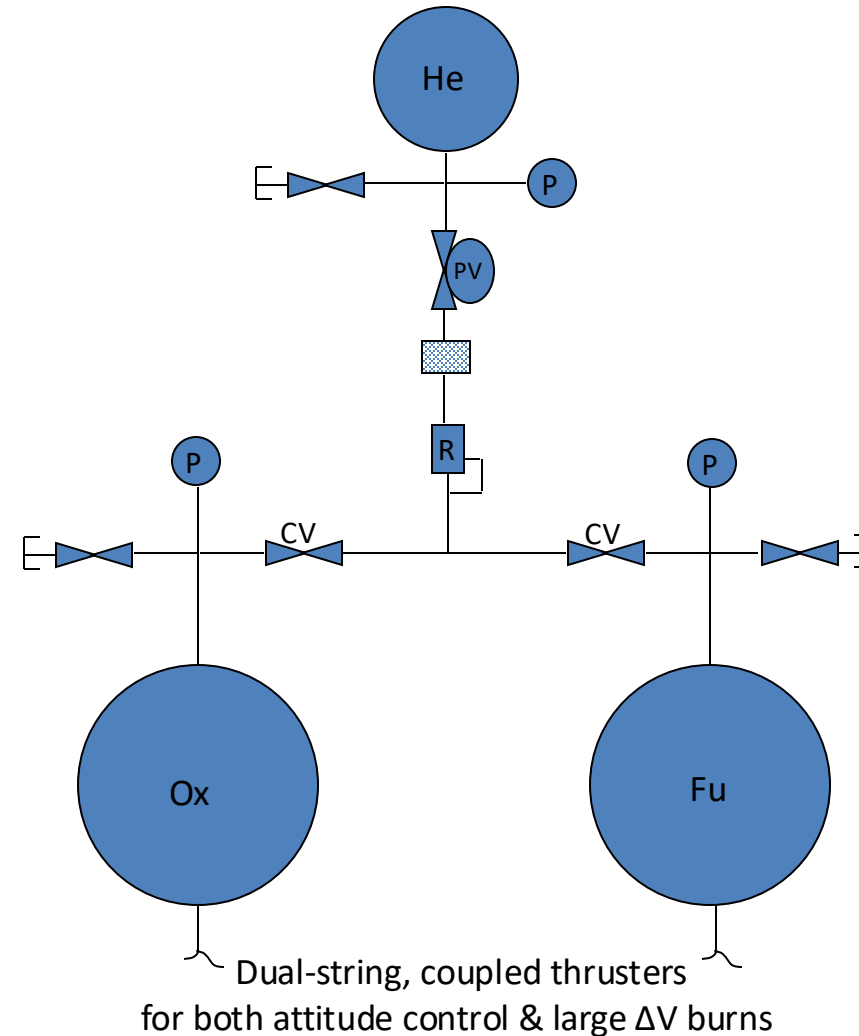
• Trade Studies

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Additional Info

- **What are the components?**
- **What type of propulsion system is this?**
 -
- **What is the mission duration?**
 - Short (≤ 1 year) versus long (> 2 years)
- **What's wrong with this design?**
 -



Interpreting Schematics

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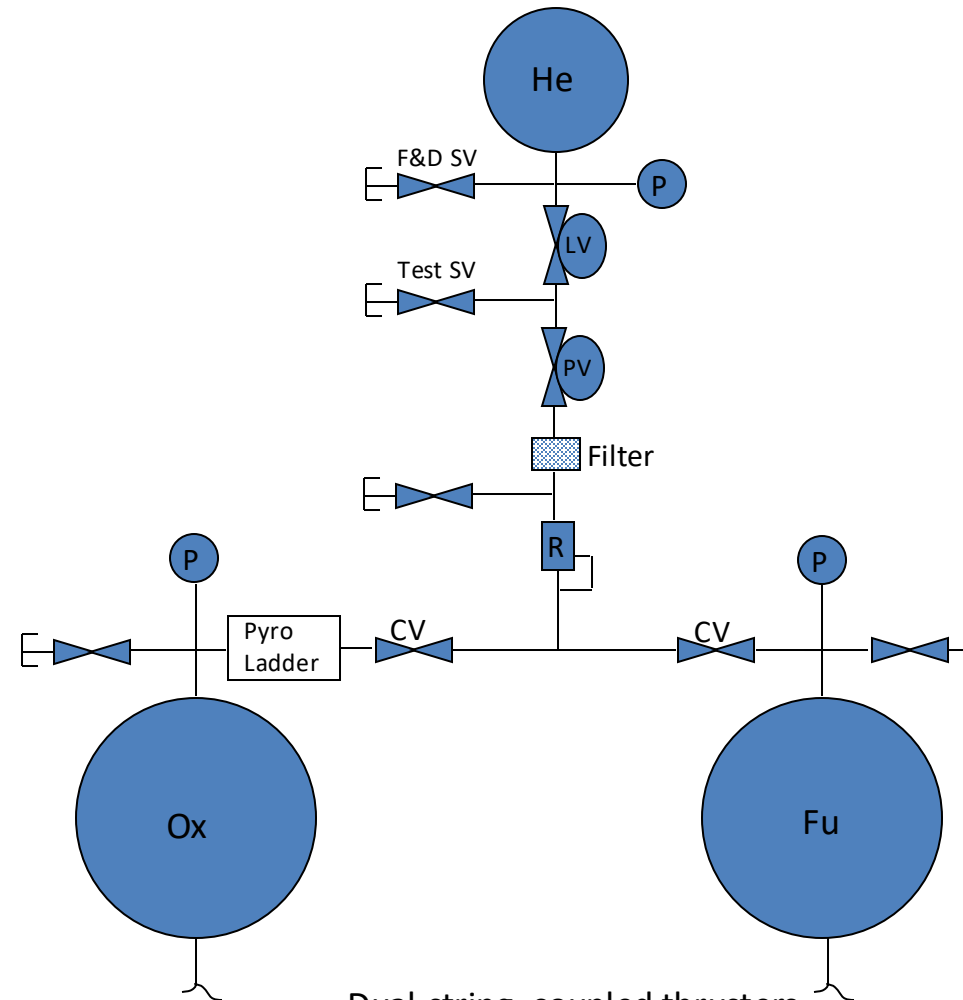
• Trade Studies

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Additional Info

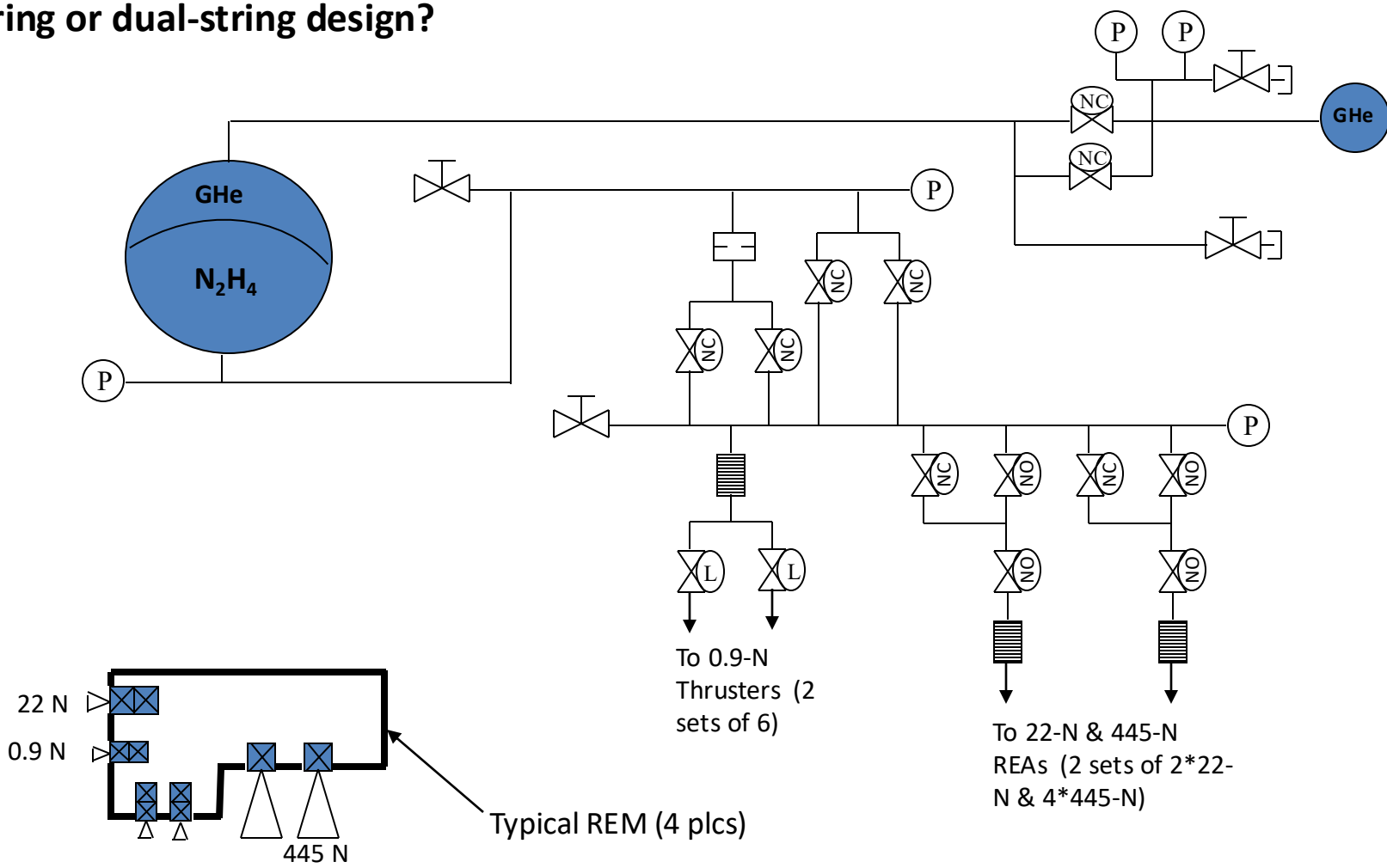
- **What are the components?**
- **What type of propulsion system is this?**
 - Regulated bipropellant
- **What is the mission duration?**
 - Short (≤ 1 year) versus long (> 2 years)
- **What's wrong with this design?**
 - No flow control (beyond regulator) after first use
 - Added high pressure latch valve
 - Insufficient control of oxidizer vapors
 - Added pyro ladder
 - Untestable regulator
 - Added service valves



Dual-string, coupled thrusters
for both attitude control & large ΔV burns

Interpreting Schematics

- How many pressurization recharges can be accomplished?
- What are the thrusters used for?
- Single string or dual-string design?



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Example 7: Magellan Venus Orbiter

More Complex Monoprop Blowdown System

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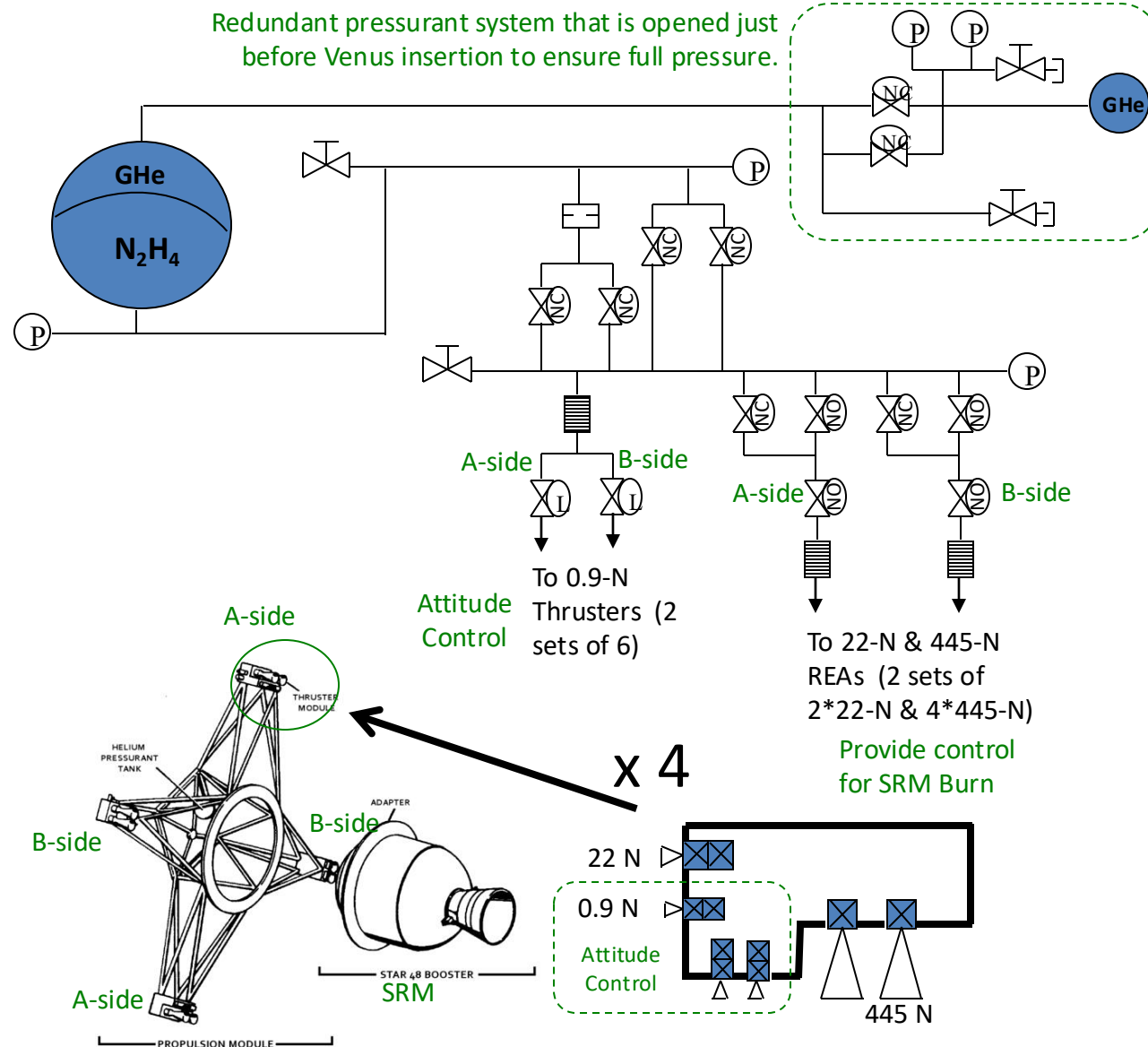
• Trade Studies

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Additional Info

- Magellan used a more complex monoprop blowdown system that incorporated an external pressurant system, 12 0.9-N thrusters for attitude control and 4 22-N & 8 445-N for orbit insertion & orbit maintenance
- The STAR 48 SRM was used for orbit insertion with the 22-N & 445-N thrusters providing additional control.
- The > 1-year flight time and redundancy requirements result in the series of pyro-valves to minimize leaks, but allowing the ability to enable/disable the individual thruster branches.
- Note that the SRM is not connected to the manifold and thus not shown (except for the graphic)



Interpreting Schematics

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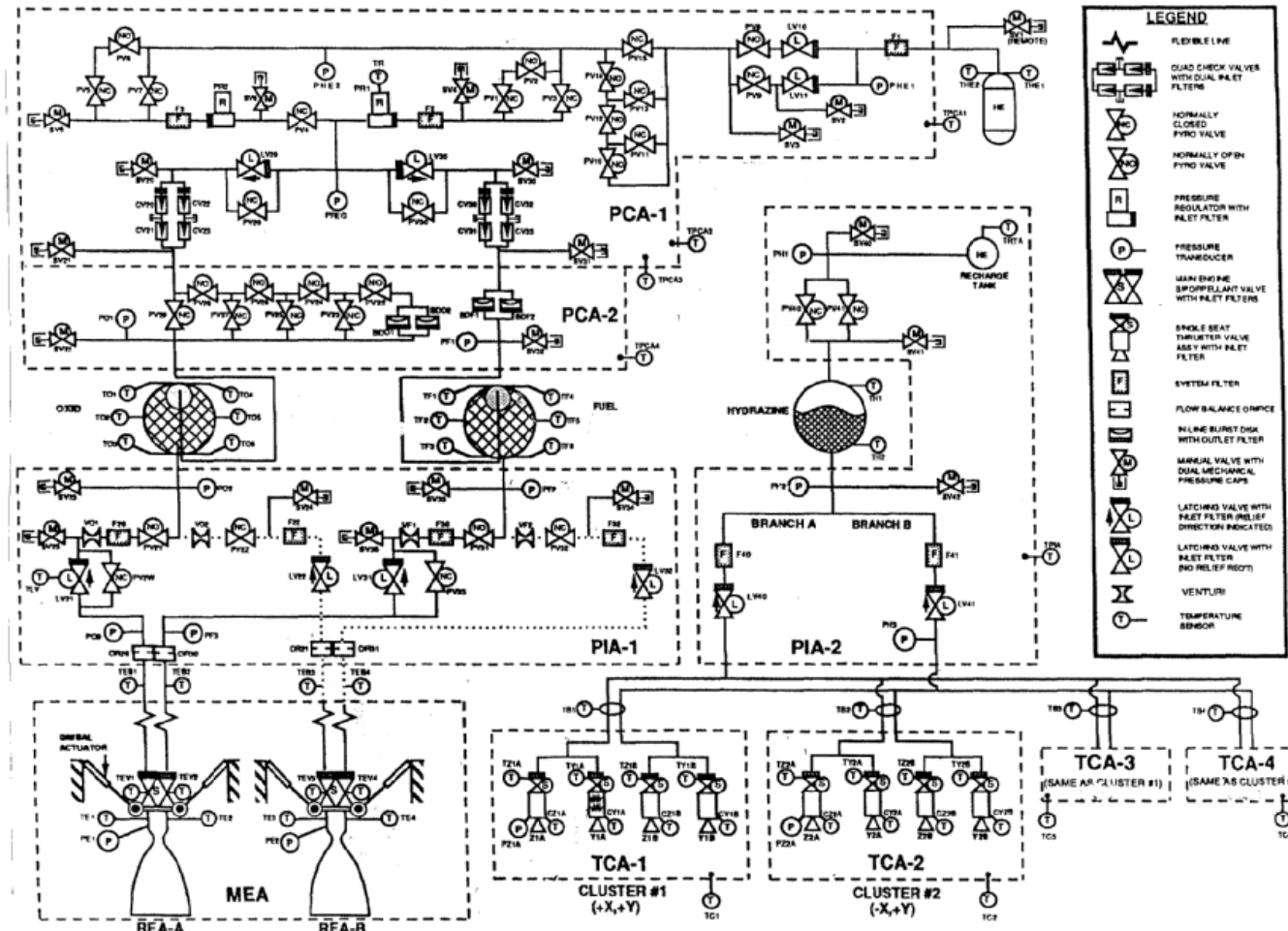
• Trade Studies

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Additional Info

- For a given area, describe the propellant / pressurant flow?
- Estimate the minimum number electrical cables/wires needed for this design?
- Describe three areas of redundancy?



Example 6: Messenger Biprop Pressurization Design

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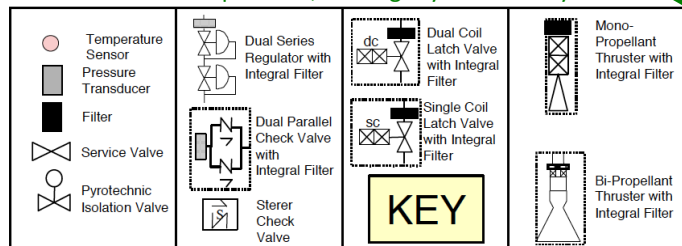
Additional Info

• Messenger executed trade study to select a pressurization system

- Constraint: 6-year trajectory with 6 large maneuvers distributed over the last three years of the cruise phase
- This long flight time & multiple maneuvers required an innovative design to minimize leaks & missing

Note the use of a pyro-ladder that allows for a discrete number of valve on/off events, but using pyros to reduce the possibility of leaking.

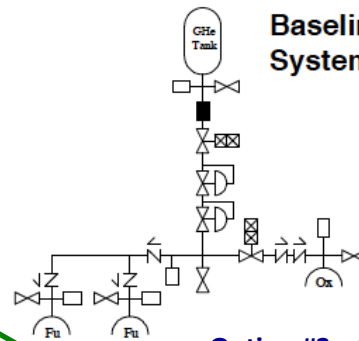
Note the same components, but slightly different symbols



Source: Wiley, S. "Design And Development Of The Messenger Propulsion System.", AIAA, 2003

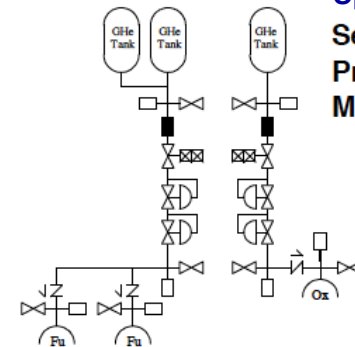
Option #1

Baseline System



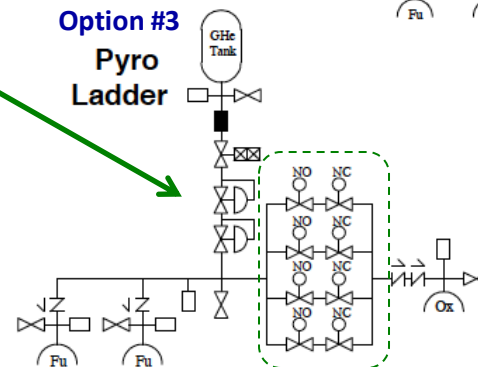
Option #2

Separate Pressurization Multiple He Tanks



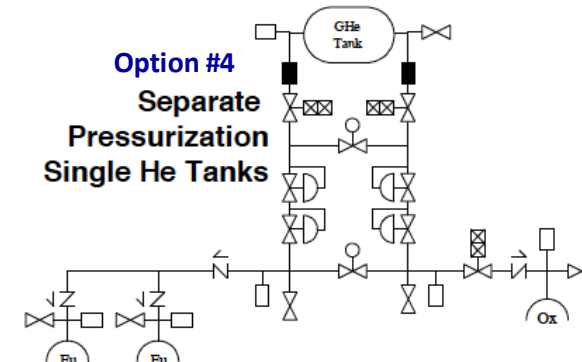
Option #3

Pyro Ladder



Option #4

Separate Pressurization Single He Tanks



Example 6: Messenger Biprop Pressurization Design

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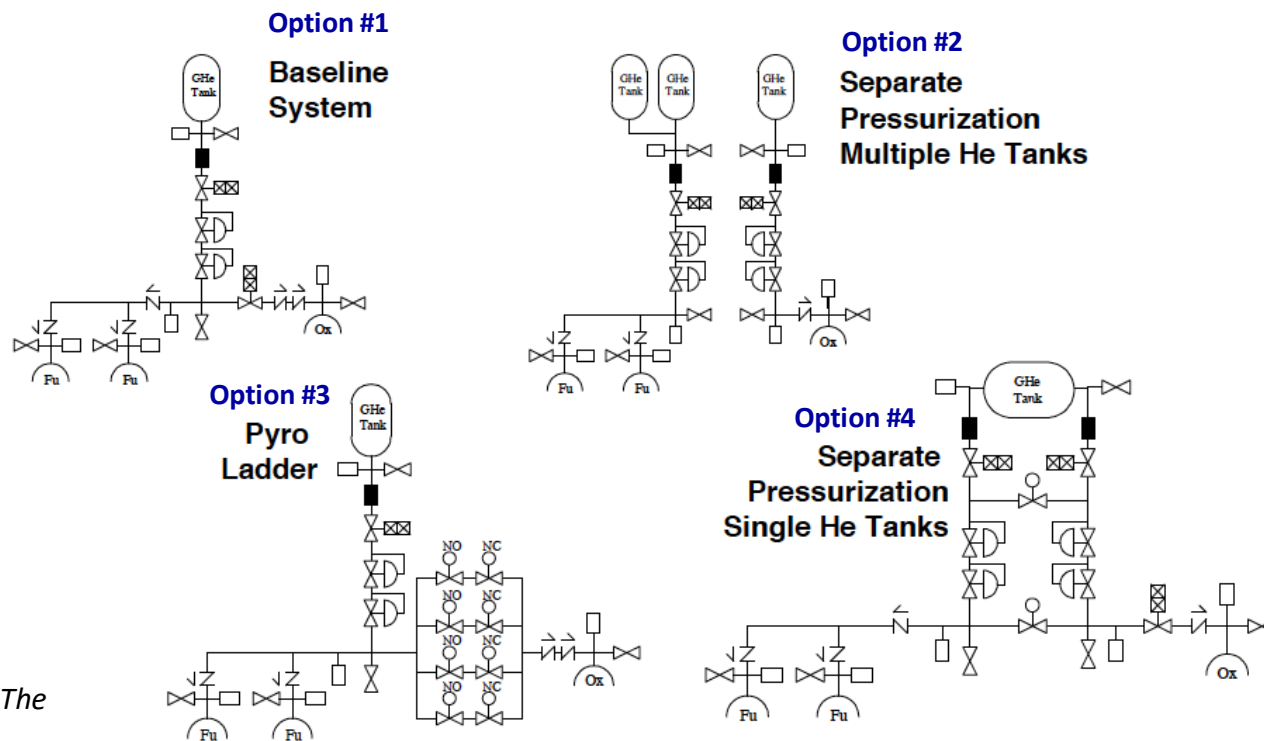
Additional Info

• Messenger executed trade study to select a pressurization system

- Constraint: 6-year trajectory with 6 large maneuvers distributed over the last three years of the cruise phase
- This long flight time & multiple maneuvers required an innovative design to minimize leaks & missing

Design Selection

- Option 1 does not provide adequate protection of mixing (Mars Observer lessoned learned)
- Options 2 & 3 are good, but result in heavier & costlier systems
- Option 4 required a slightly new type of tank design, but was more efficient and safer.



Source: Wiley, S. "Design And Development Of The Messenger Propulsion System.", AIAA, 2003

Propulsion System Trades (1 of 2)

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Additional Info

	Pros	Cons
Solid Rocket Motor	• Simple design (few/no moving parts) • Minimal pre-launch preparation • Easy to get ready to operate on-orbit • No leakage/slosh concerns • Least total system mass for certain range of total impulse • Highly reliable solution for kick stage (fixed total impulse) applications	• Inadvertent ignition or explosion, potentially catastrophic • Inflexible for propellant mass changes, once cast and machined • Complex, toxic exhaust gases can be problematic for some applications
Electric Propulsion	• Very high efficiency for high ΔV trajectories • Minimizes overall flight system mass	• Requires significant power (eg, kW)
Cold Gas	• Lowest total system mass and cost for low Isp • Generally simpler than liquid design • Simple pre-launch preparation (non-hazardous fluid) • Easy to prepare for on-orbit operation • No slosh/cg issues	• Leakage concerns, especially for long-life missions • Late increases in required total impulse can quickly result in non-optimum system

Propulsion System Trades (2 of 2)

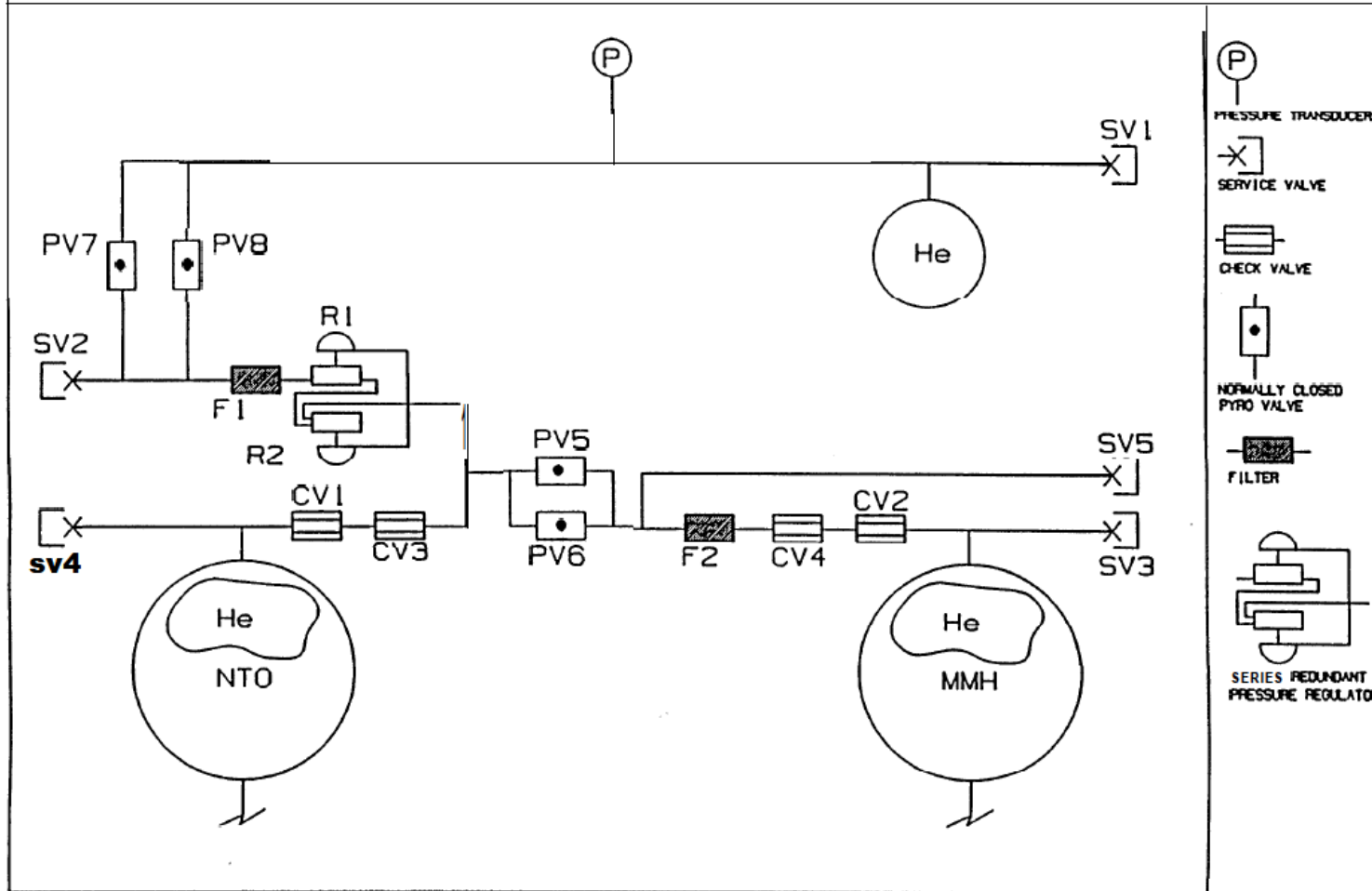
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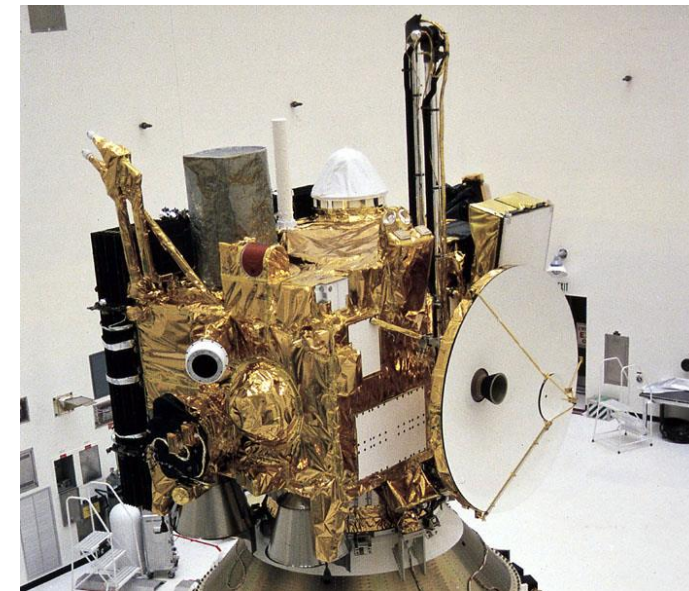
	Pros	Cons
Monopropellant	• Simple, reliable blowdown system • Reasonable mass-efficient choice for small/medium ΔV systems • Variety of heritage tanks & thrusters available • Very stable operation over a wide range of impulse bits and burn durations • Small thrusters qualified for > 1M pulses • Stable operation for very long-life missions • Simple set of exhaust products	• Some tank mass inefficiency w/blowdown • Regulated monoprop systems more mass efficient but added cost and complexity • Hydrazine is toxic (fueling) • Not mass-efficient for very small total impulse systems • More expensive than a cold gas system
Bipropellant	• High specific impulse results in good mass efficiency for medium/large ΔV systems • Biprop mixture ratio equates to equal volume fuel/oxidizer tanks for commonality • Variety of heritage tanks & thrusters available • Stable operating over a range of press/temp • Long-burns (> 60 min) demonstrated	• No qualified thrusters for low thrust applications (<10N) • Requires regulated pressurization system • Exhaust products much more complex and toxic than monoprop • Bipropellants are toxic (fueling) • System more complex than monoprop systems, with resultant higher cost and lower reliability
Dual-Mode (Mono + Biprop)	• Higher performance over biprop; good mass efficiency for medium/large systems • Simpler hardware layout (vs. having separate systems) • Complexity comparable to biprop systems	• Less stable operating range versus biprop • Mixture ratio is results in different size fuel & oxidizer tanks • More limited selection of heritage thrusters

Case Study: Mars Observer

BI-PROPELLANT PRESSURIZATION SUB-SYSTEM SCHEMATIC



- JPL launched Mars Observer on 9/25/92 to study the Martian...
 - Surface
 - Atmosphere
 - Climate
 - Magnetic field
- Three days prior to orbit insertion, communication was lost and never re-established.



Propulsion

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